Foundations of Neuroscience (FoN)

Session 1 - Lecture

Neurons, Synapses and the Role of Neurotransmitters

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HUAZHONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Learning Outcomes

- Classification of neurons
- Synaptic classifications – chemical and electrical
- Basic aspects of neurotransmission
  - Critical processes of neurotransmission
  - Neurotransmitter receptors
- Post synaptic effects of receptor activation – ionotropic and metabotropic signalling mechanisms

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Question: What are the processes that a human body have to take in order to jump?

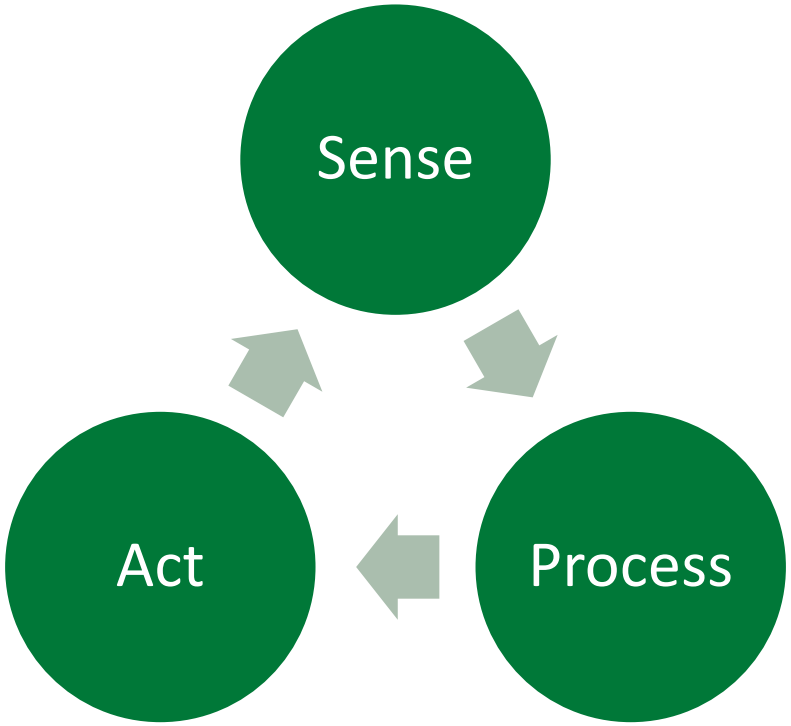


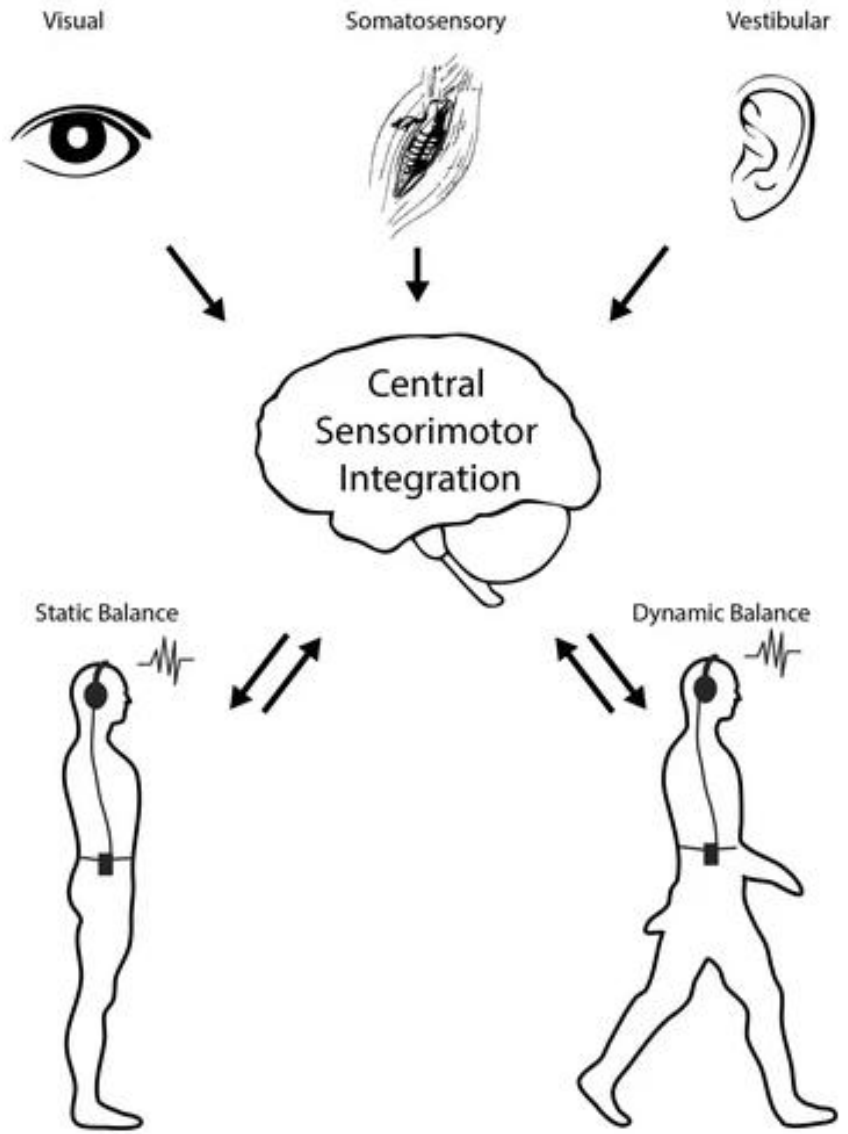
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Steps required: Sensing, processing and executing movement

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The sensorimotor feedback loop



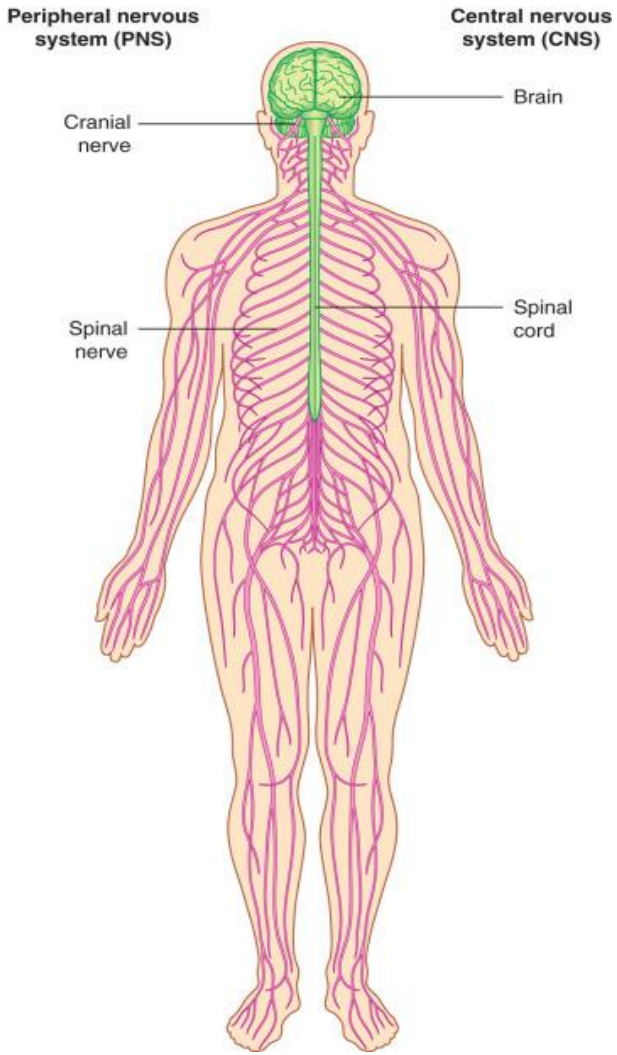


Core component of the body that regulates all the steps is the **nervous system**

Fino, P.C., Peterka, R.J., Hullar, T.E. *et al. BMC Neurol* **17**, 41 (2017).

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The nervous system



Central nervous system

- The brain and spinal cord

Peripheral nervous system

- **Somatic** nervous System (muscles)
  - Sensory (**afferent**: to the brain)
  - Motor (**efferent**: from the brain)
- **Autonomic** nervous system (organs)
  - **Sympathetic** (fight or flight)
  - **Parasympathetic** (rest and digest)
  - **Enteric** (blood flow, mucosal secretions, immunity and endocrine)

Special senses

- Sight, smell, hearing, taste

Somatic = Voluntary

Further details in Somatosensory System Lecture

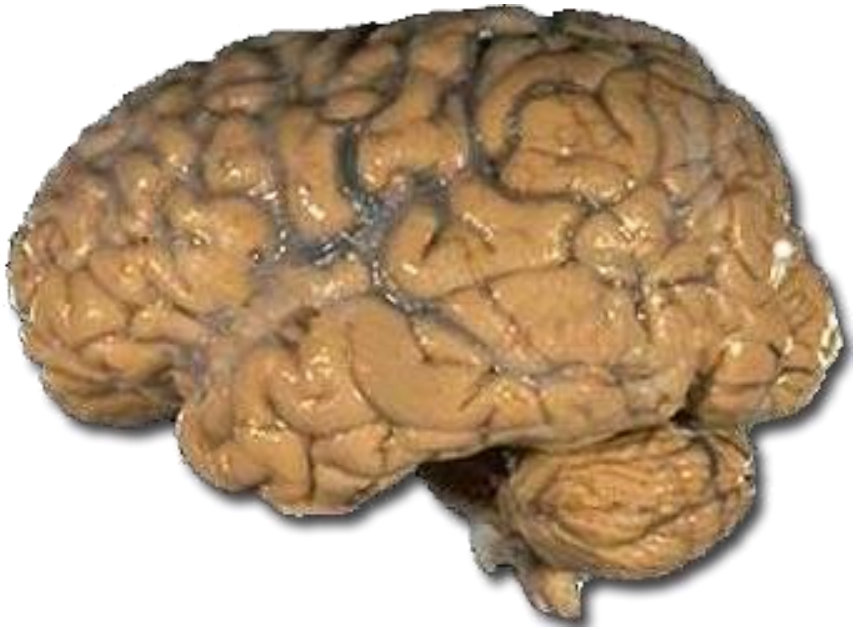
Autonomic = Involuntary/ Housekeeping

Further details in Autonomic Nervous System Lecture

Further details in Vision, Hearing, Taste and Smell Lectures

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The human brain



- One hundred billion neurons
- 150,000-180,000 km nerve fibres
- 0.15 quadrillion synapses in cortex (150 thousand billion)

- 1 neuron lost every second
- or ~85,000 /day (~31 million/year)

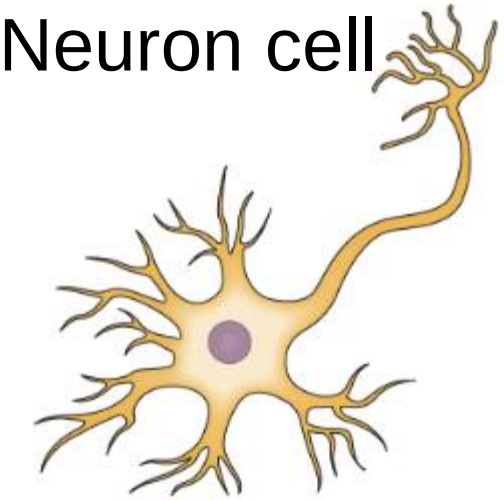
NIH image of human brain
License: http://www.nih.gov/about/Faqs.htm#copyright

Further details in Brain, Spinal Cord and Peripheral Nerves Tutorial

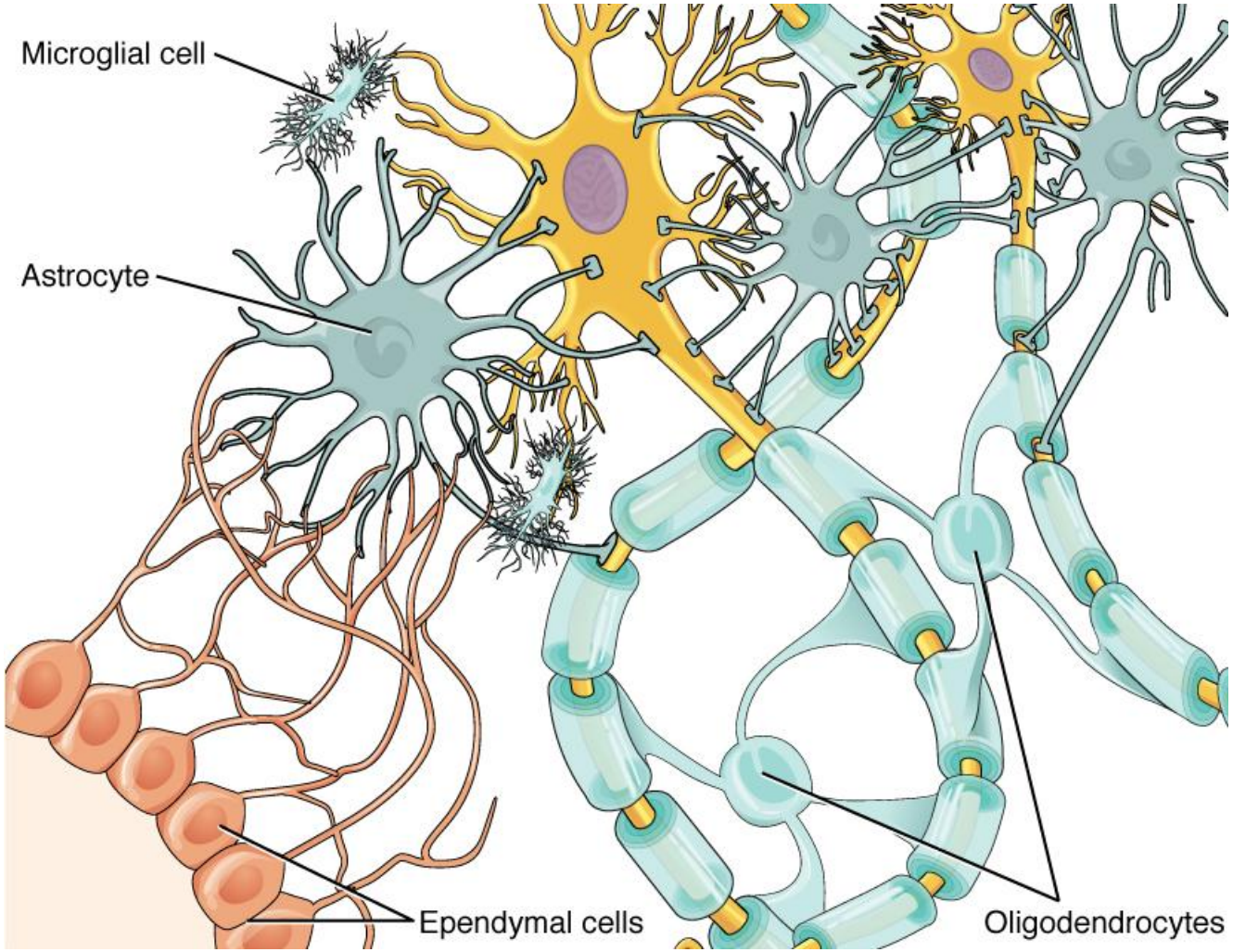
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Neuron and glial cells – primary cells in the brain

Neuron cell



- Neurons are responsible for signalling
- Glial cells (Microglial cells, Astrocyte, Oligodendrocytes, Ependymal cells) are support cells
- Glial cells are ~10x more abundant



Glial cells

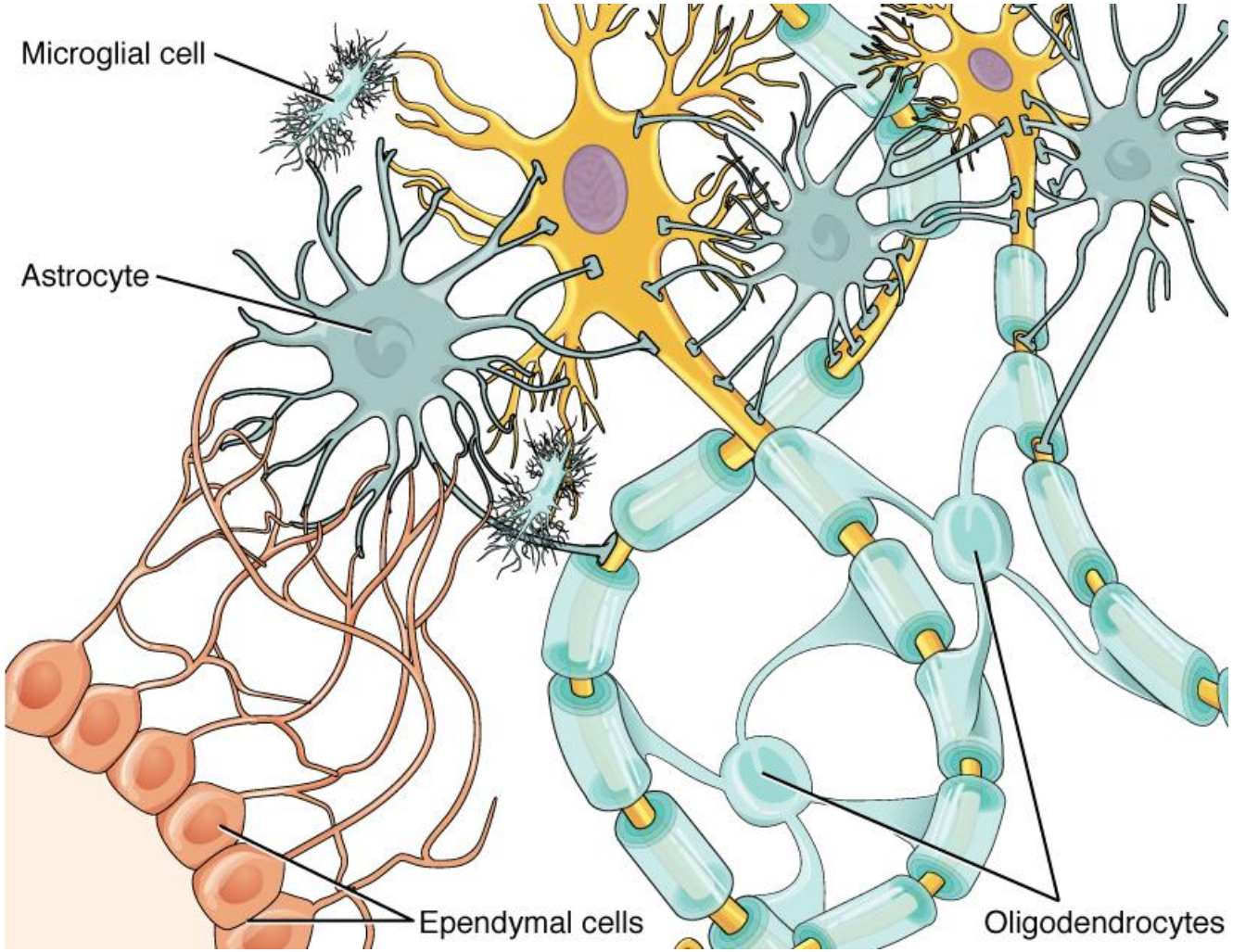
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bio.libretexts.org

Function of Glial cells

Glial cell	Function
Astrocyte	Metabolic support, maintaining blood brain barrier, maintain chemical balance in the brain
Oligodendrocyte	Insulation and myelination to improve signal transmission
Microglia	Immune surveillance and phagocytosis
Ependymal cell	Creating cerebral spinal fluid

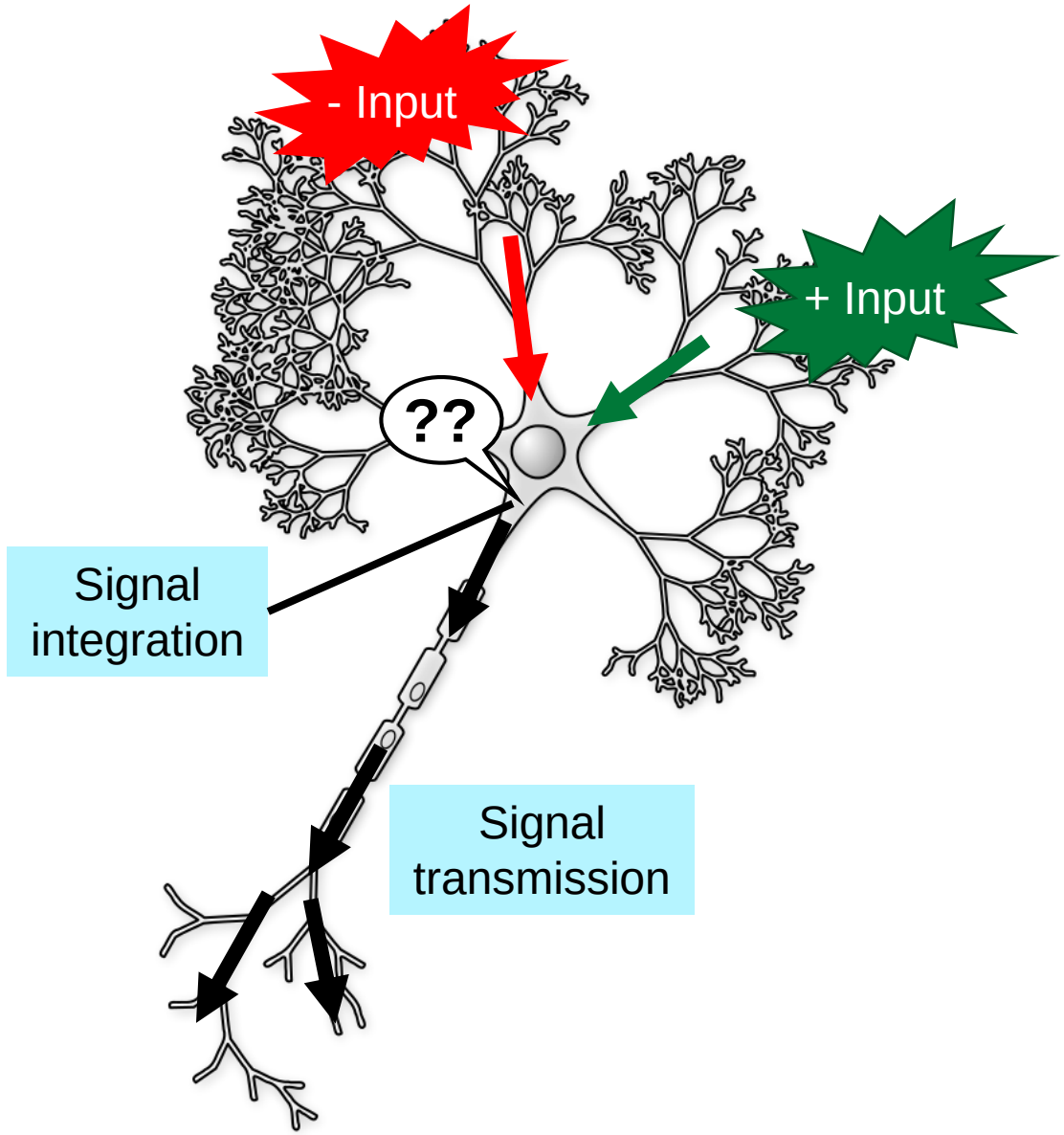
Glial cells create and maintain the optimal physical and chemical environment that allows neurons to survive, communicate, and adapt.



Further details in Glia: Form and Function Lecture

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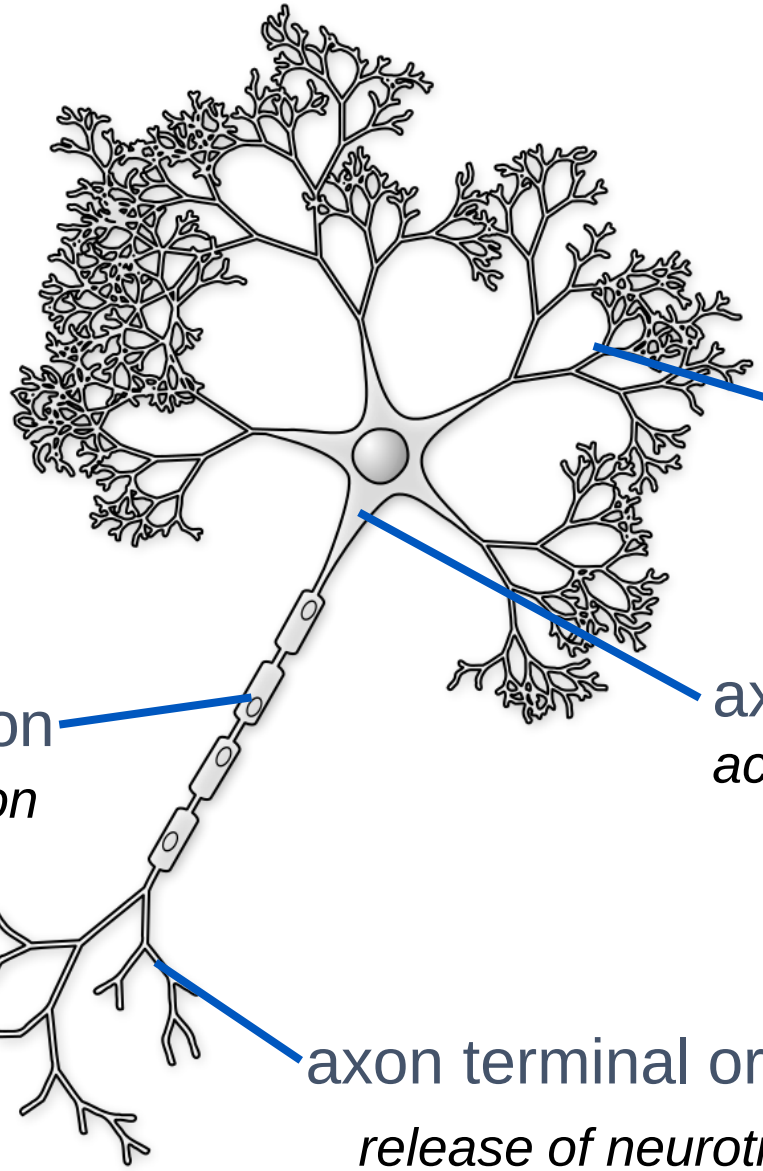
The Neuron



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Neuronal specialisations for communication



Dendrite is derived from the Greek word dendron, meaning "tree."

dendrites

input from other neurons

axon hillock

action potential generation

axon

impulse conduction

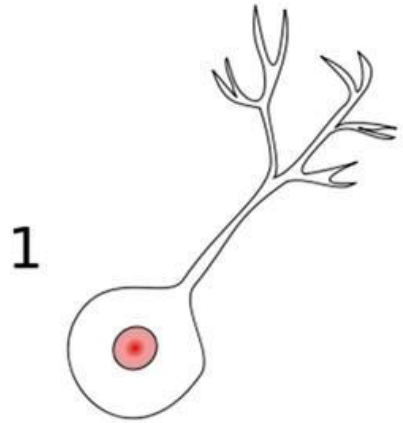
axon terminal or synaptic bouton

release of neurotransmitter

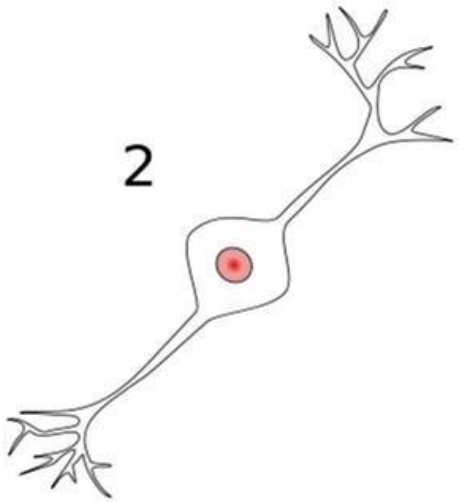
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Classification of neurons



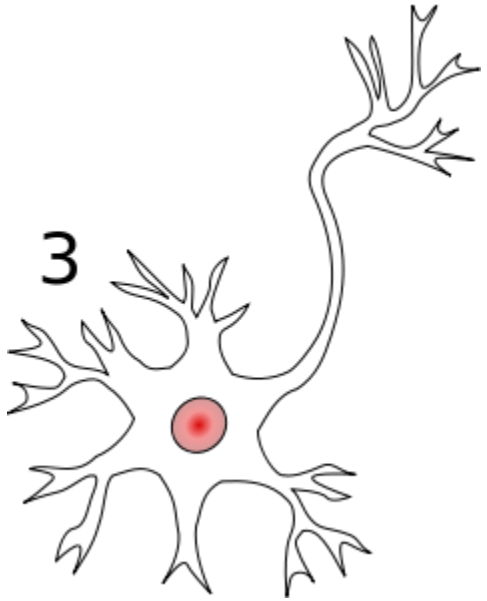
Unipolar cell
Single axonal process
Rare in humans
e.g. invertebrate neuron



Bipolar cell
Two axonal processes
Special **sensory relay** with minimal integration of signals
e.g. retinal bipolar cell

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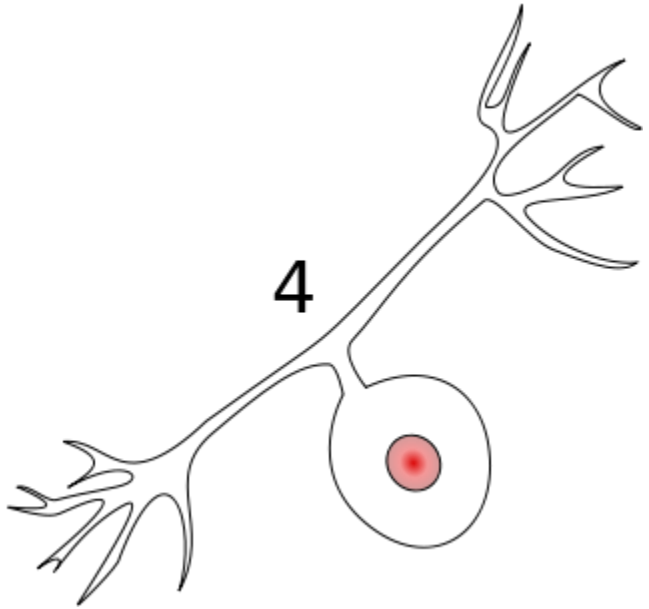
Classification of neurons



Multipolar cell

Multiple axonal processes
Most common neuron found in humans
Defined by having **one axon** and **many dendrites**

e.g. Spinal motor neuron Purkinje cell of cerebellum
Pyramindal cell of hippocampus



Pseudo-unipolar cell

Single axonal process
Act as a relay station to for signals to “pass-through”
e.g. dorsal root ganglia cell
(primary sensory neuron)

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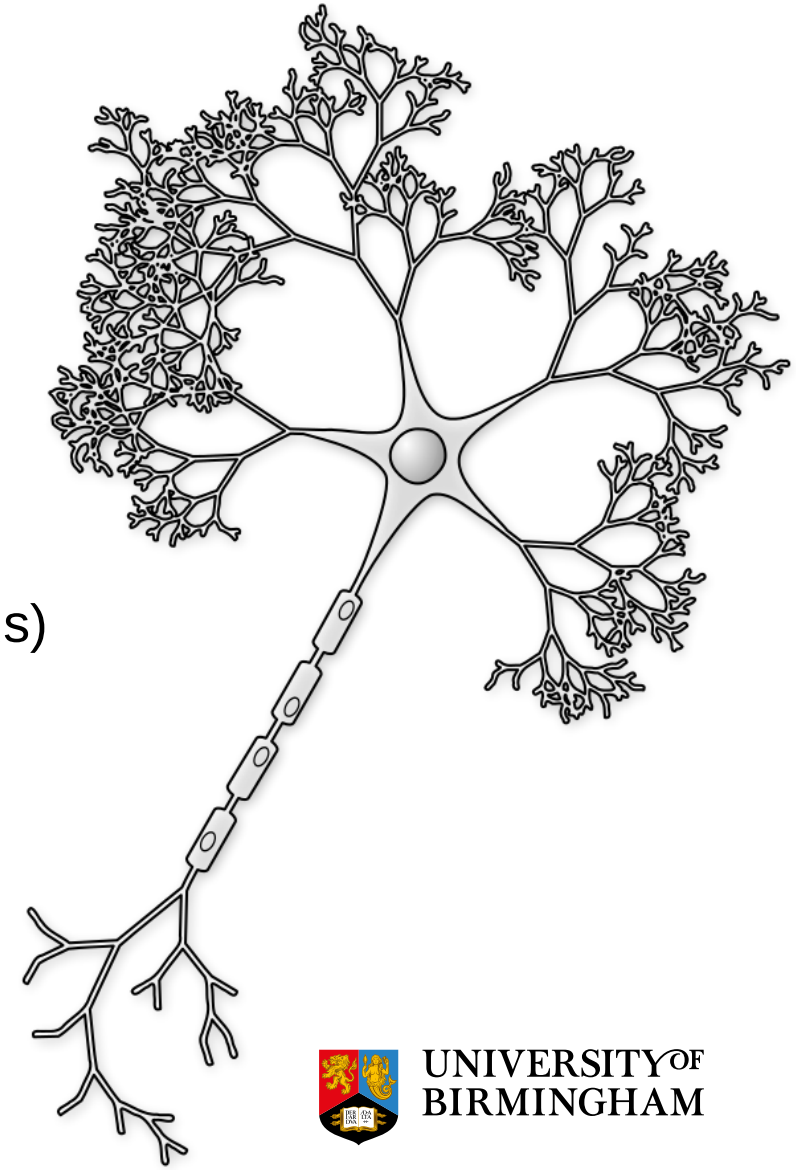
Classification of neurons

Neurons can be classified on the basis of a number of morphological criteria:

- Number of major processes: uni- bi- multi-polar

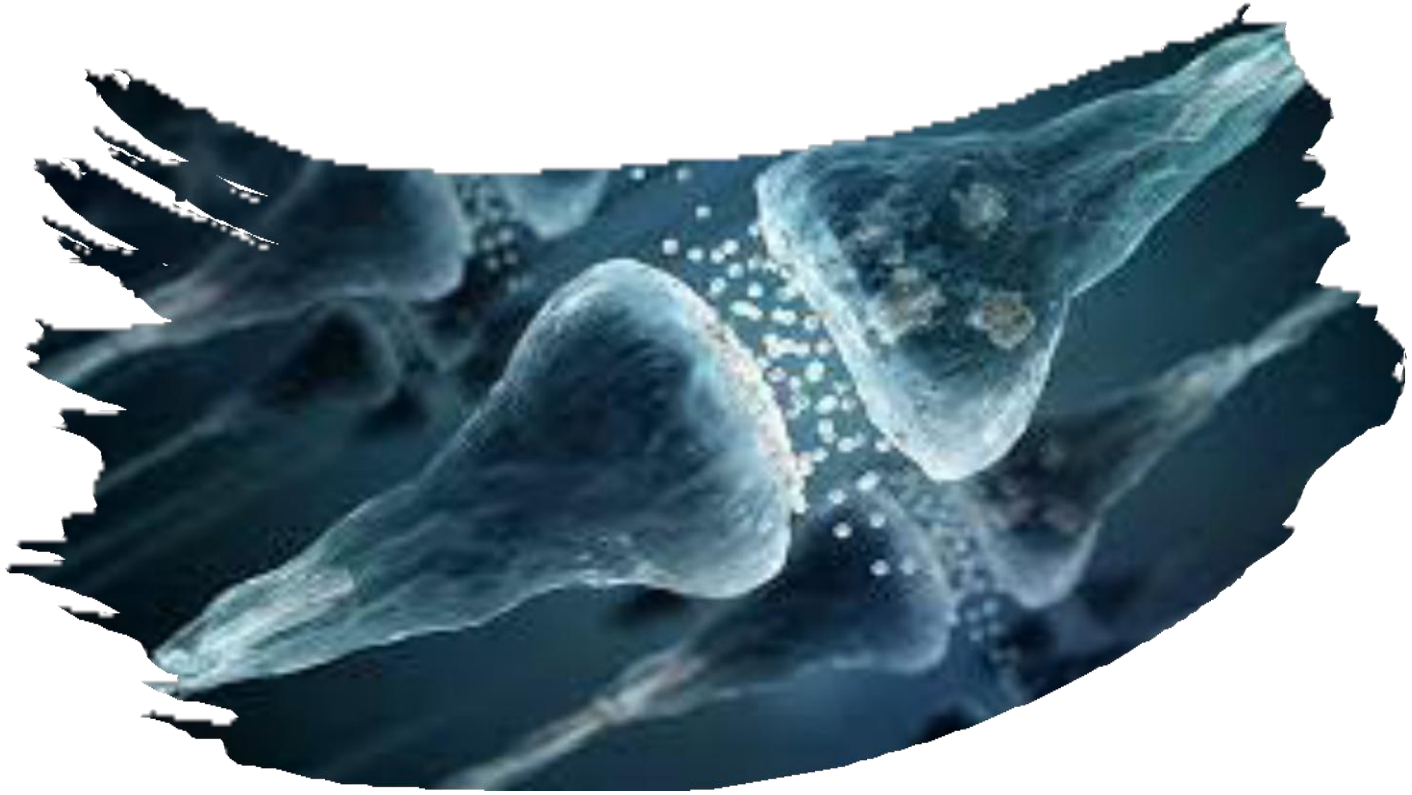
- Dendrites:
  - shape of tree (*e.g.* stellate, pyramidal)
  - presence/absence of ‘spines’ (spinous or aspinous)

- Connections: sensory, motor, interneurons
- Axon length: Golgi type I (projection) Golgi type II (local circuit)
- Neurotransmitter: cholinergic, GABAergic



Further details in Neuronal Circuits Tutorial

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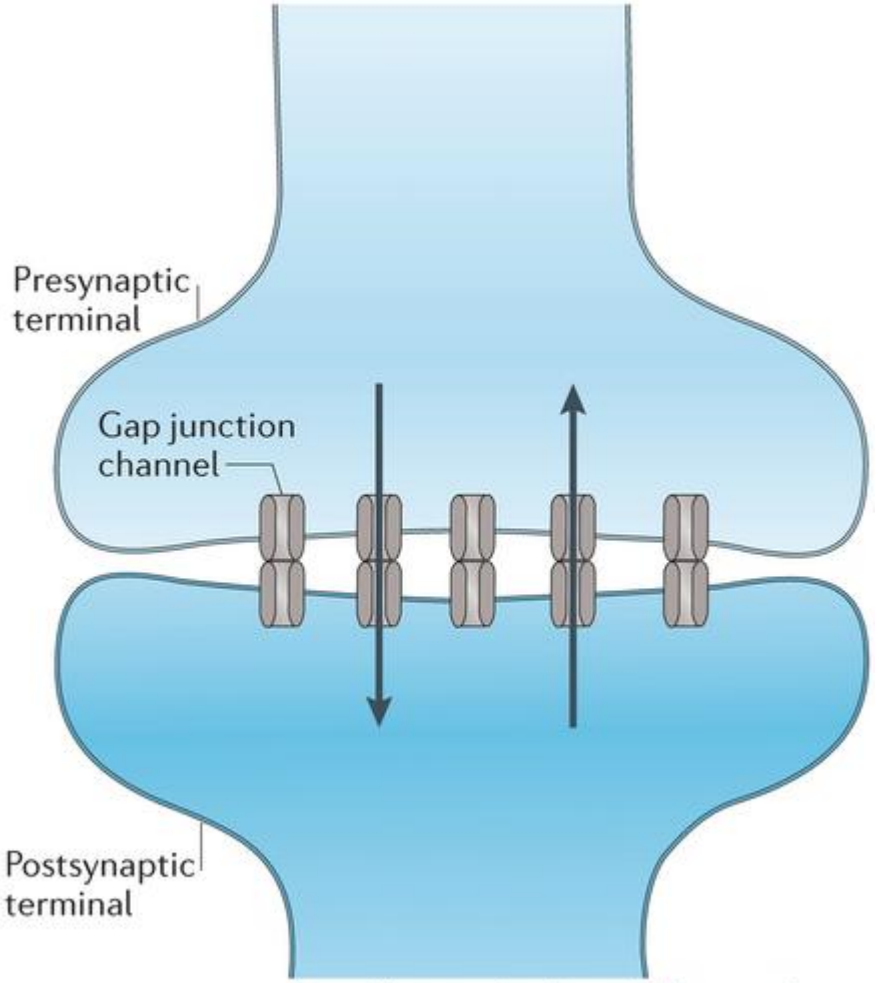
Synapses

Points of communication between a neurone and a post- synaptic neurone

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Electrical Synapse / Gap Junction

- Fastest and most primitive
- Bi-directional transfer of information
- Allows synchronous activity between neurons
- Relatively rare between neurons in the CNS, although important in development and in heart muscles
- Glia-neuron, glia-glia communication (also cardiac myocytes)

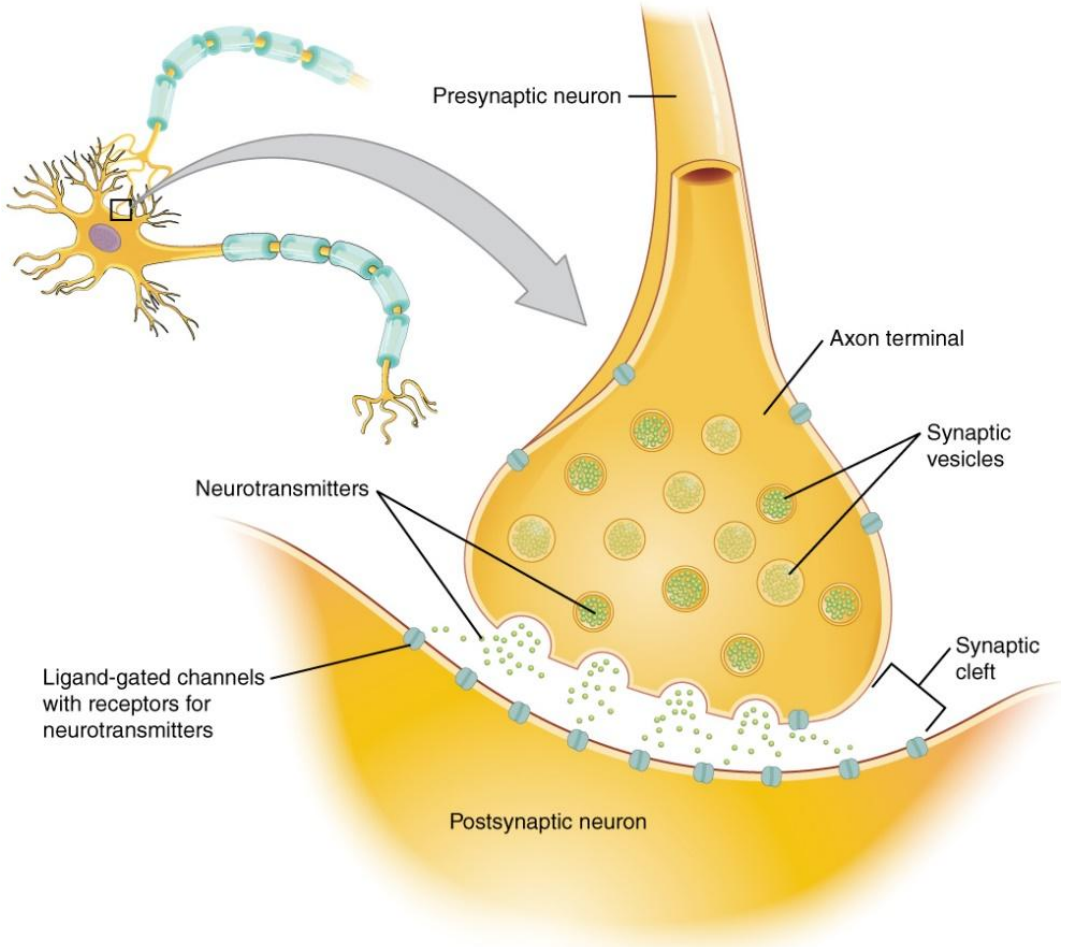


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Chemical Synapse

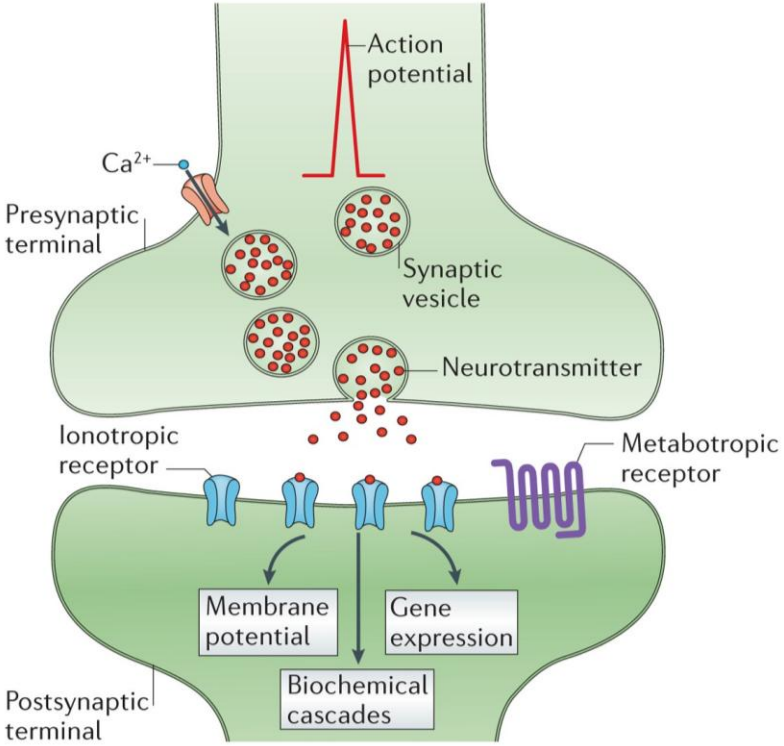
- Electrical activity within the presynaptic neuron results in the release of chemical messengers known as **neurotransmitters**
- Neurotransmitters pass through a chemical synapse (**synaptic cleft**)
- Neurotransmitters bind to the receptors present on the **postsynaptic** neuronal membrane.
- Useful and essential when the signal is passed over complex and large routes; **unidirectional** transfer of information



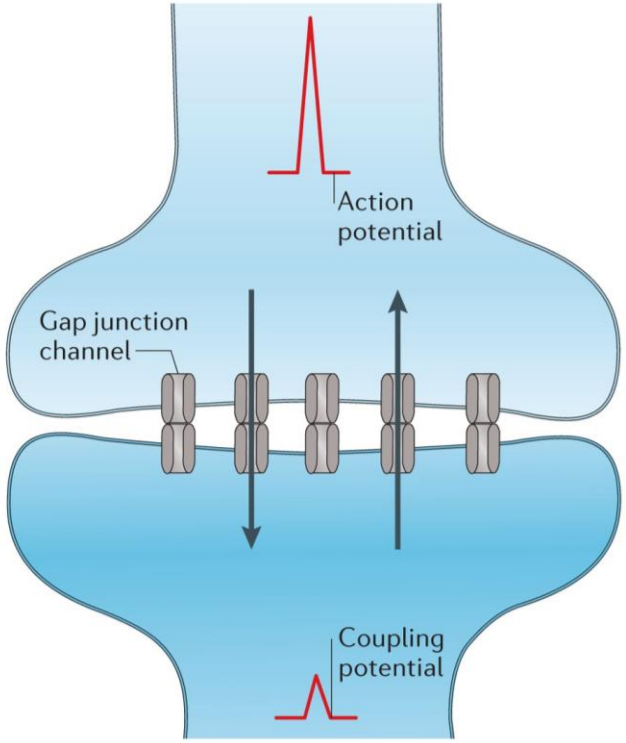
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Types of Synapses

Chemical synapse



Electrical synapse

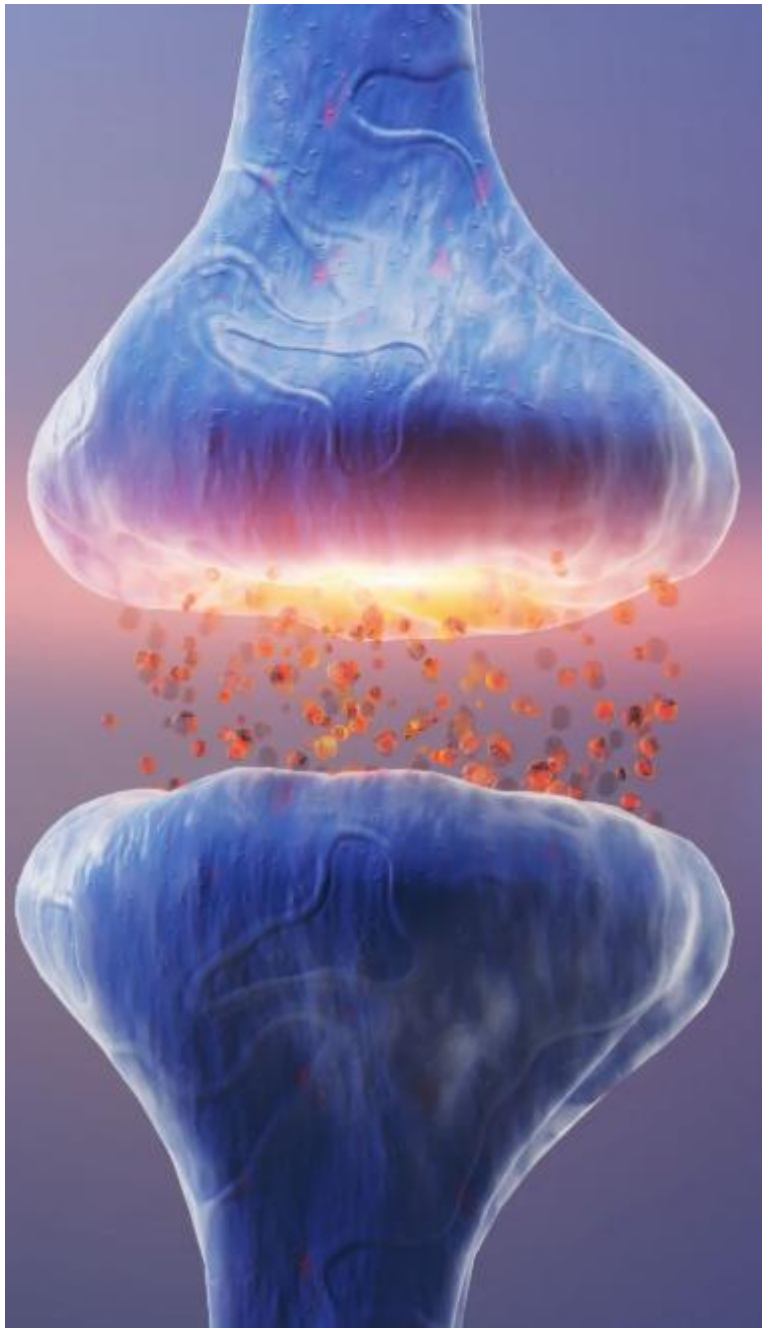


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Feature	Chemical Synapse	Electrical Synapse
Mechanism of transmission	Uses **neurotransmitters**	Uses direct ion flow through **gap junctions**
Signalling transmission	Across a physical gap called **synaptic cleft**	Direct connection between cytoplasm via **gap junctions**
Speed of transmission	Slower	Extremely fast
Direction of signalling	**Unidirectional** (one-way)	**Bidirectional** (two-way)
Prevalence	Common in human brain	Common in heart muscle
Fatigue	Can fatigue due to neurotransmitter running out	Does not fatigue easily

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Further details in Synaptic Signalling lecture



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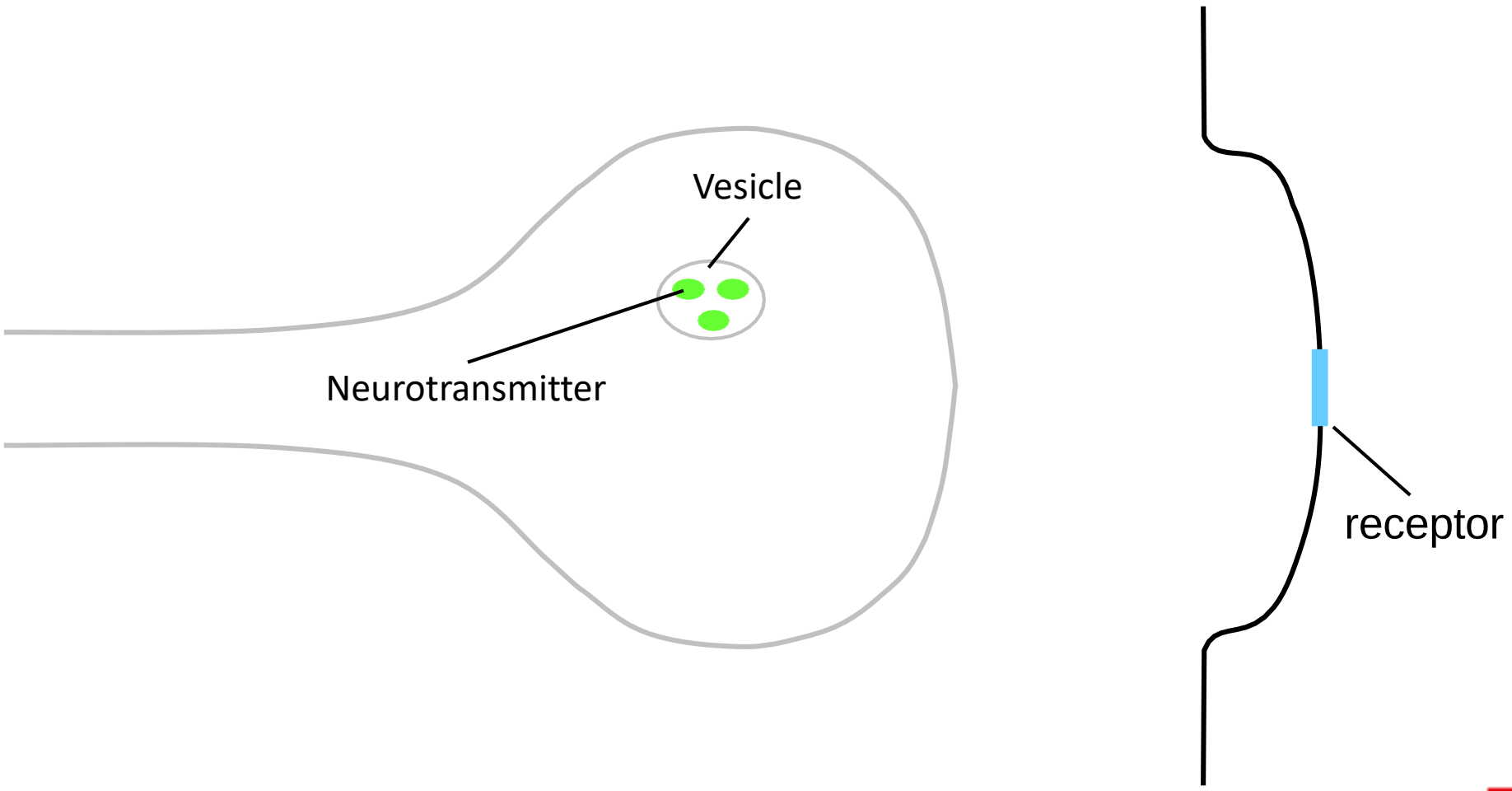
Neurotransmission

Neurotransmission is the process by which one neuron communicates with another neuron, a muscle cell, or a gland cell by releasing **chemical messengers** called **neurotransmitters** across a gap called the **synapse**.

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Presynaptic terminal

Postsynaptic terminal

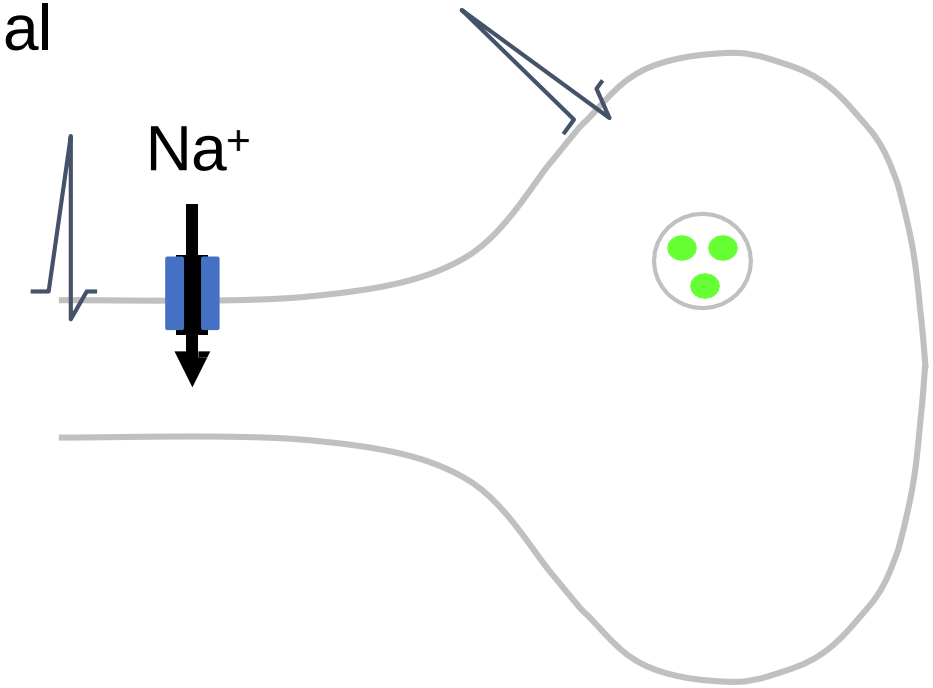


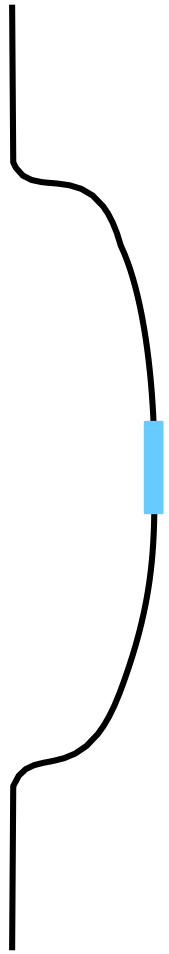
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Presynaptic terminal

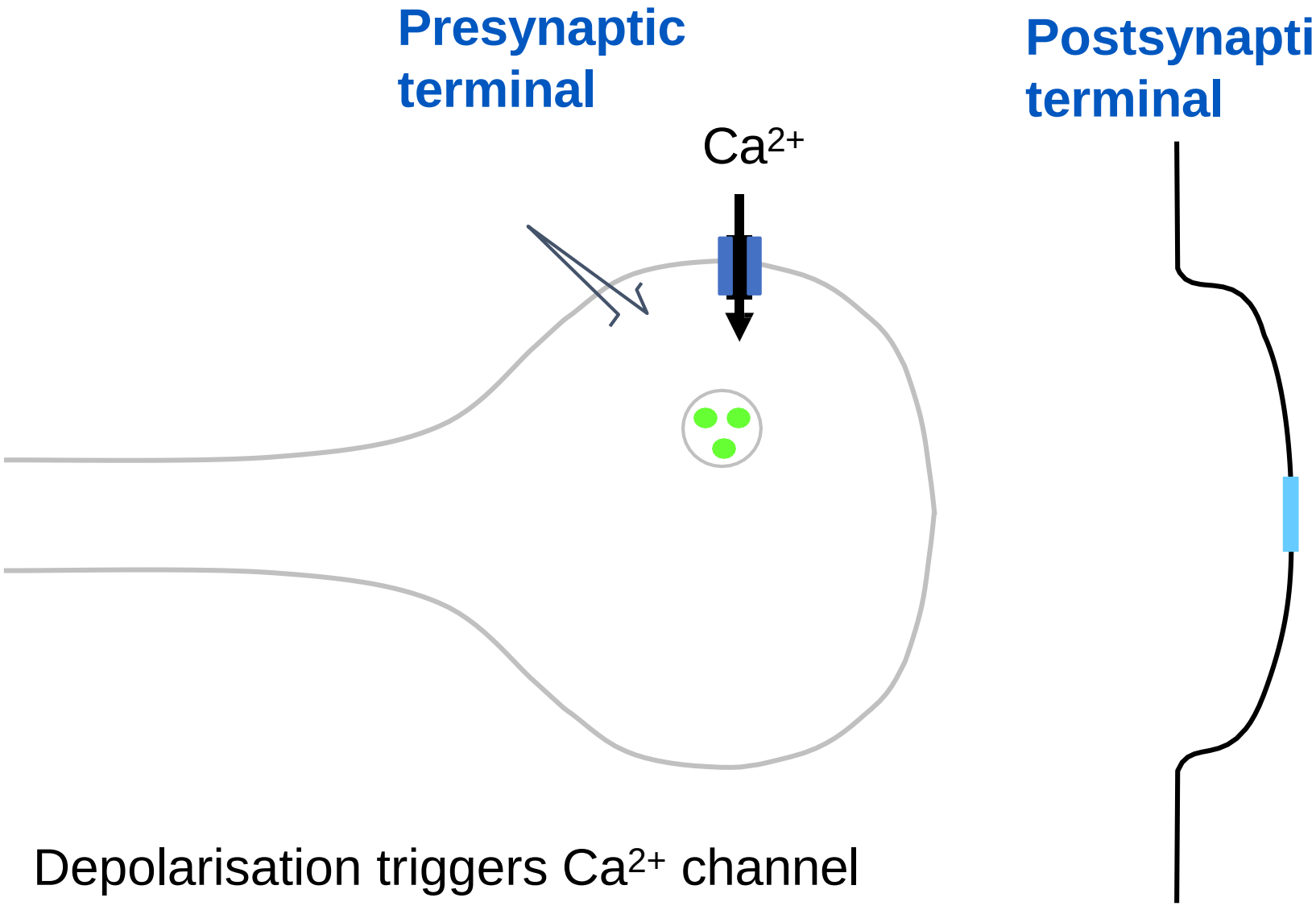
Postsynaptic terminal

action potential invades nerve terminal



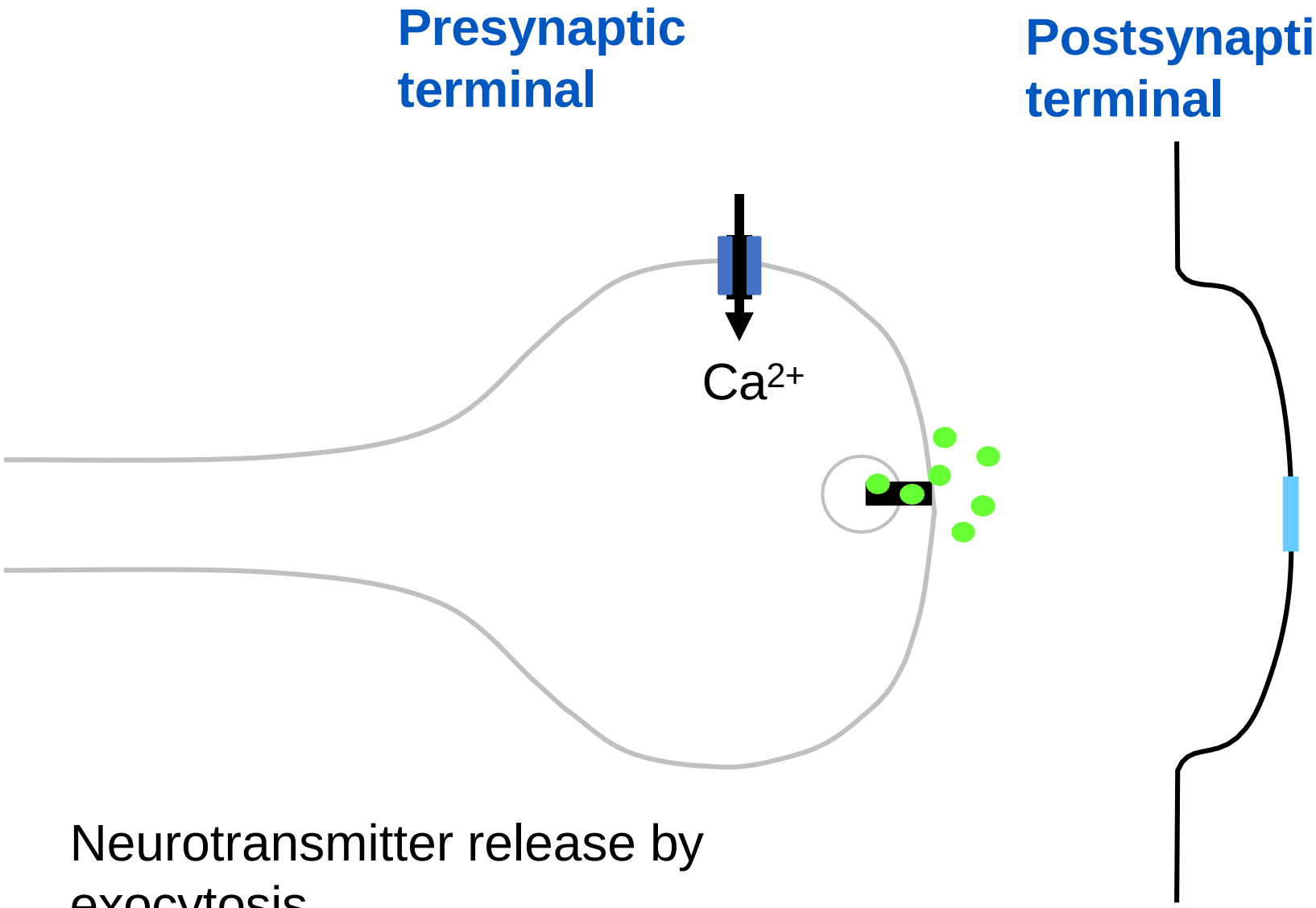


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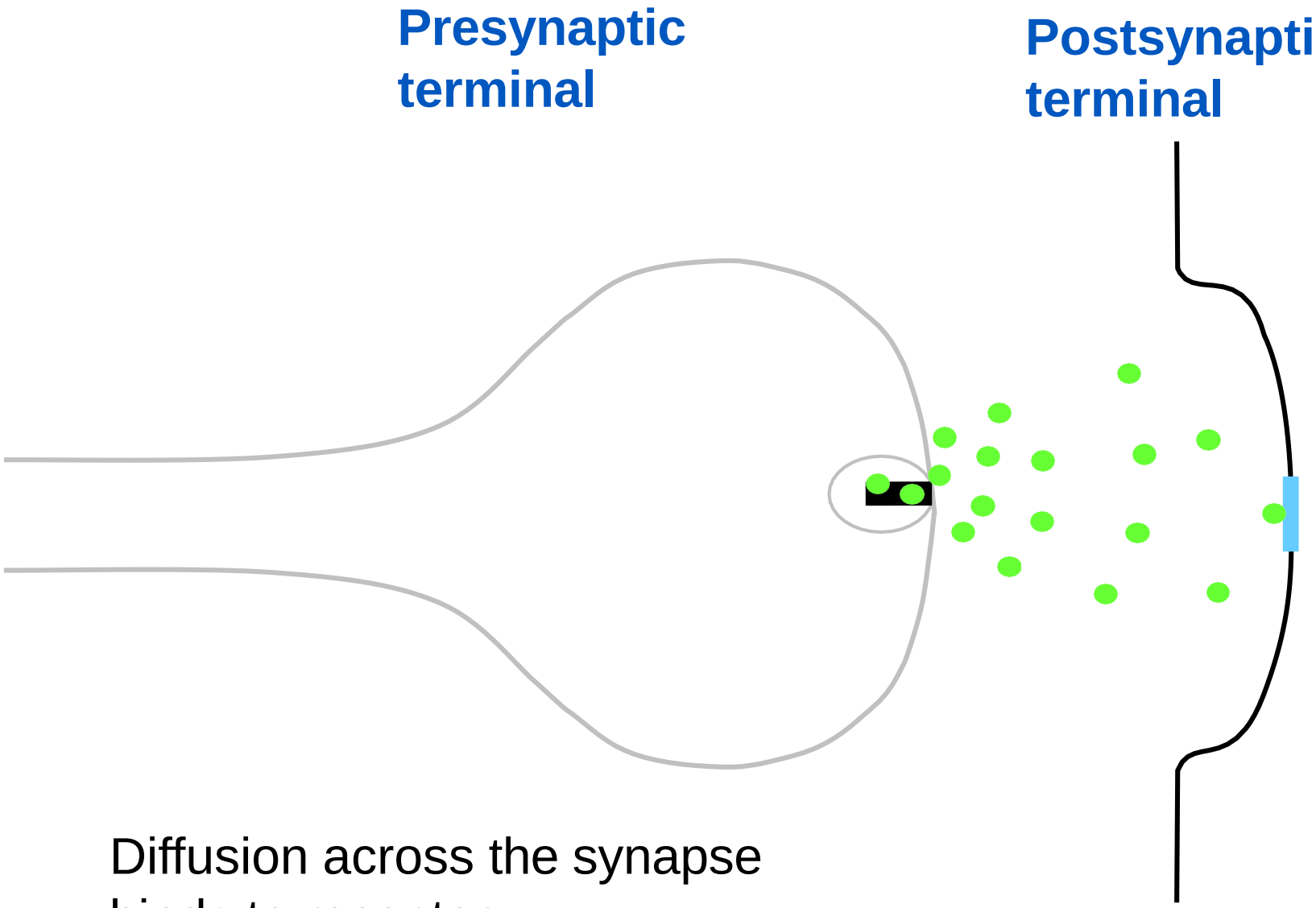
Depolarisation triggers Ca^{2+} channel opening $\Rightarrow$ Ca^{2+} influx

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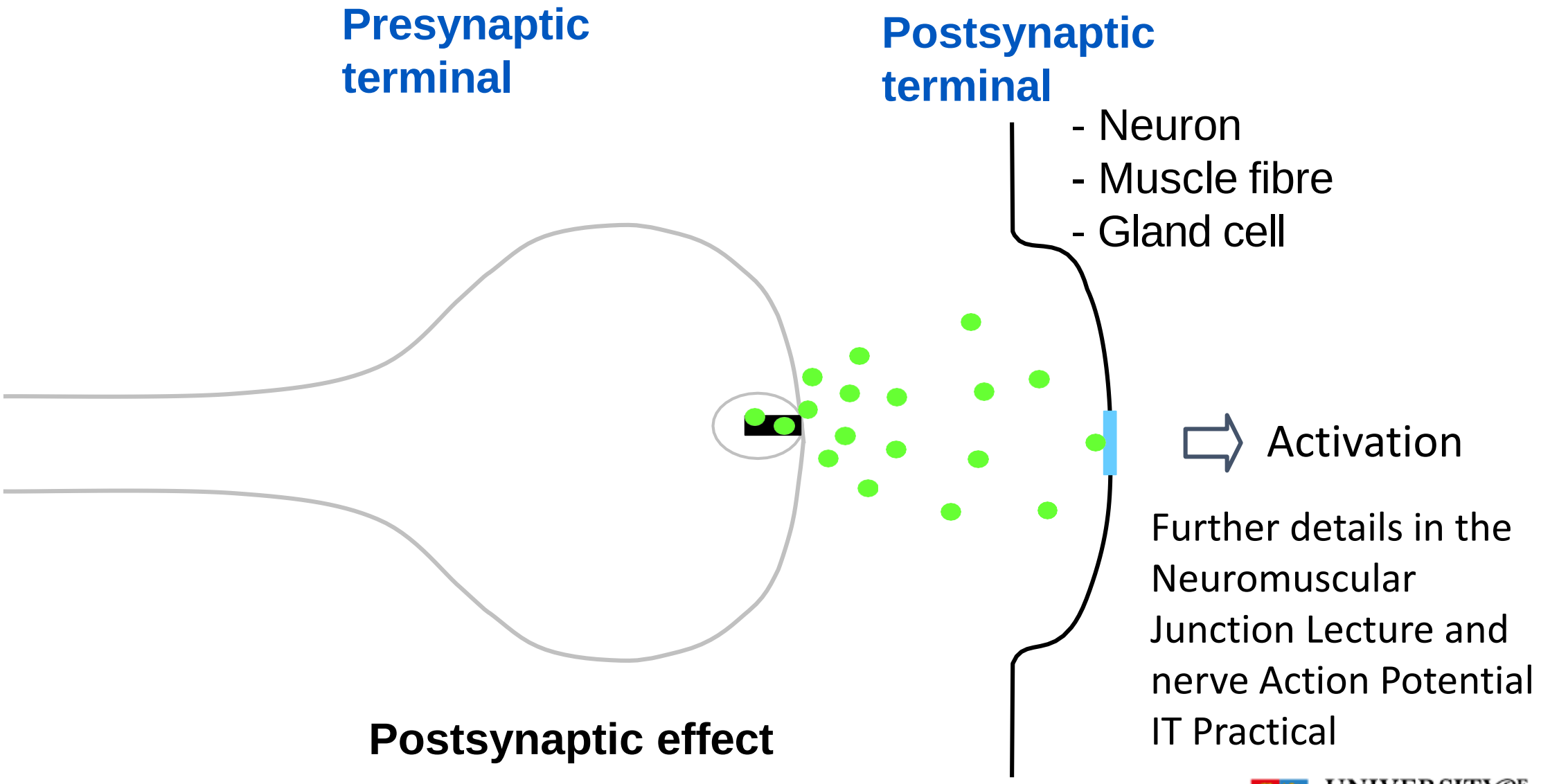
Neurotransmitter release by exocytosis

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Diffusion across the synapse binds to receptor

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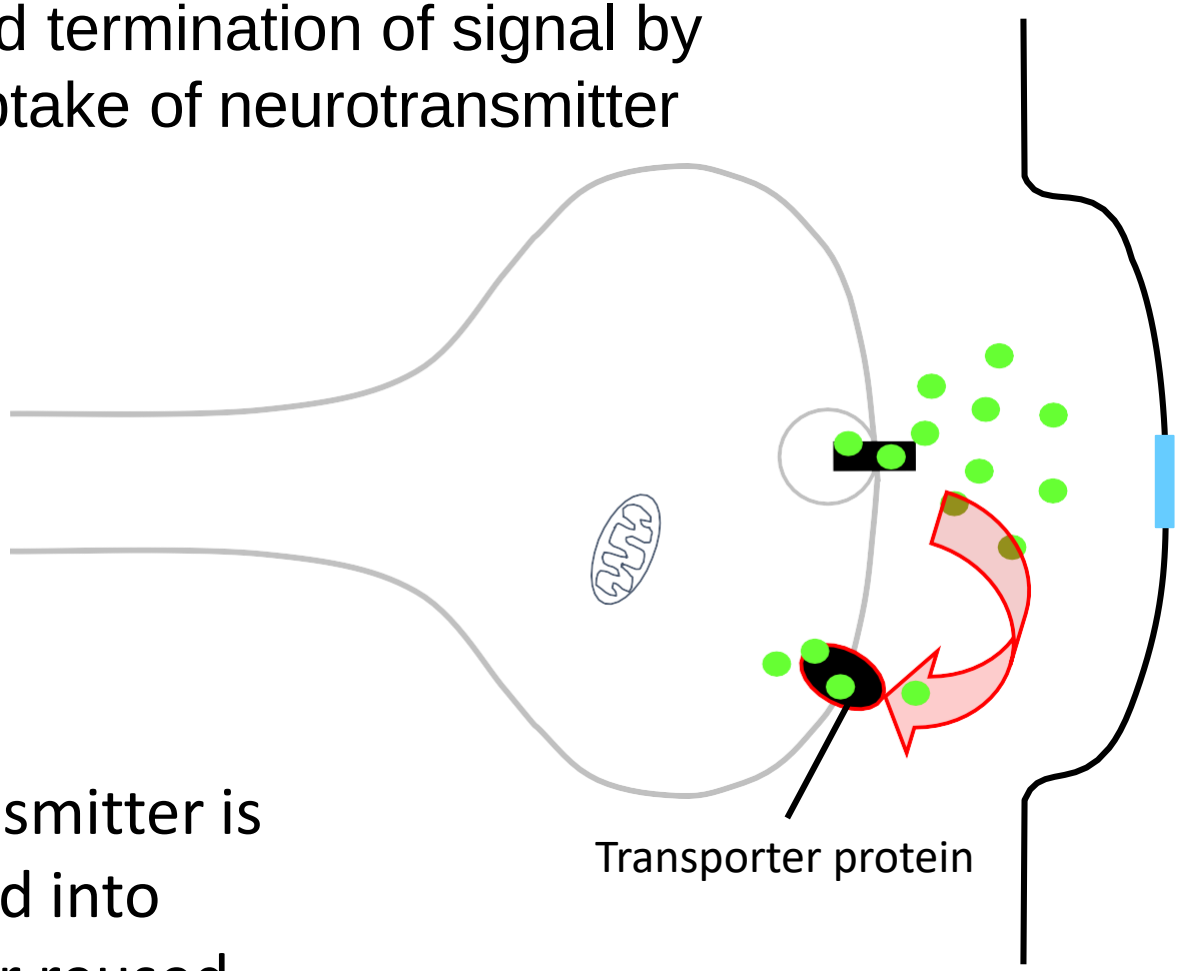


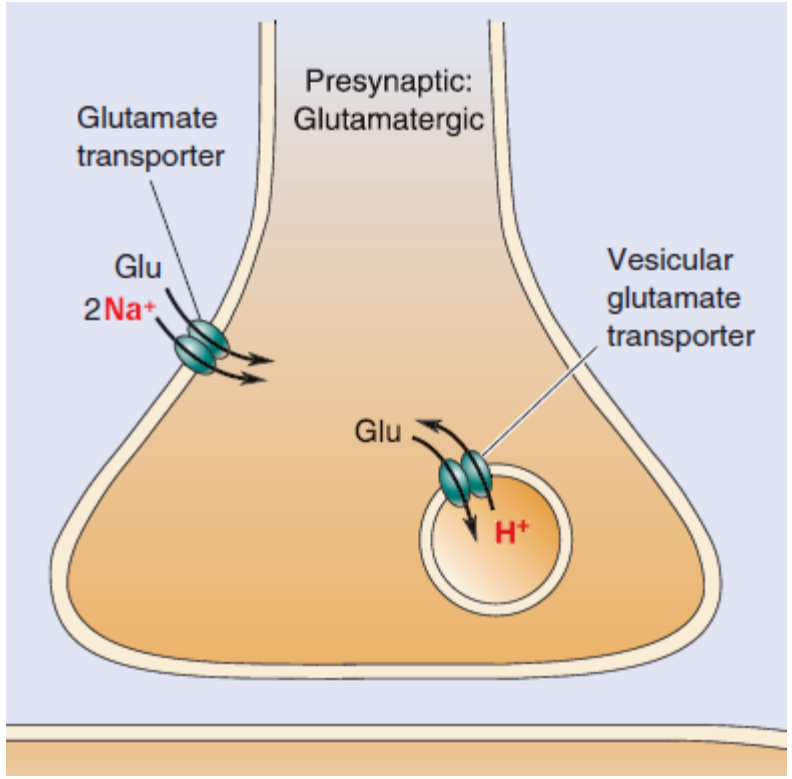
Further details in the Neuromuscular Junction Lecture and nerve Action Potential IT Practical

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Signal termination

Rapid termination of signal by re-uptake of neurotransmitter



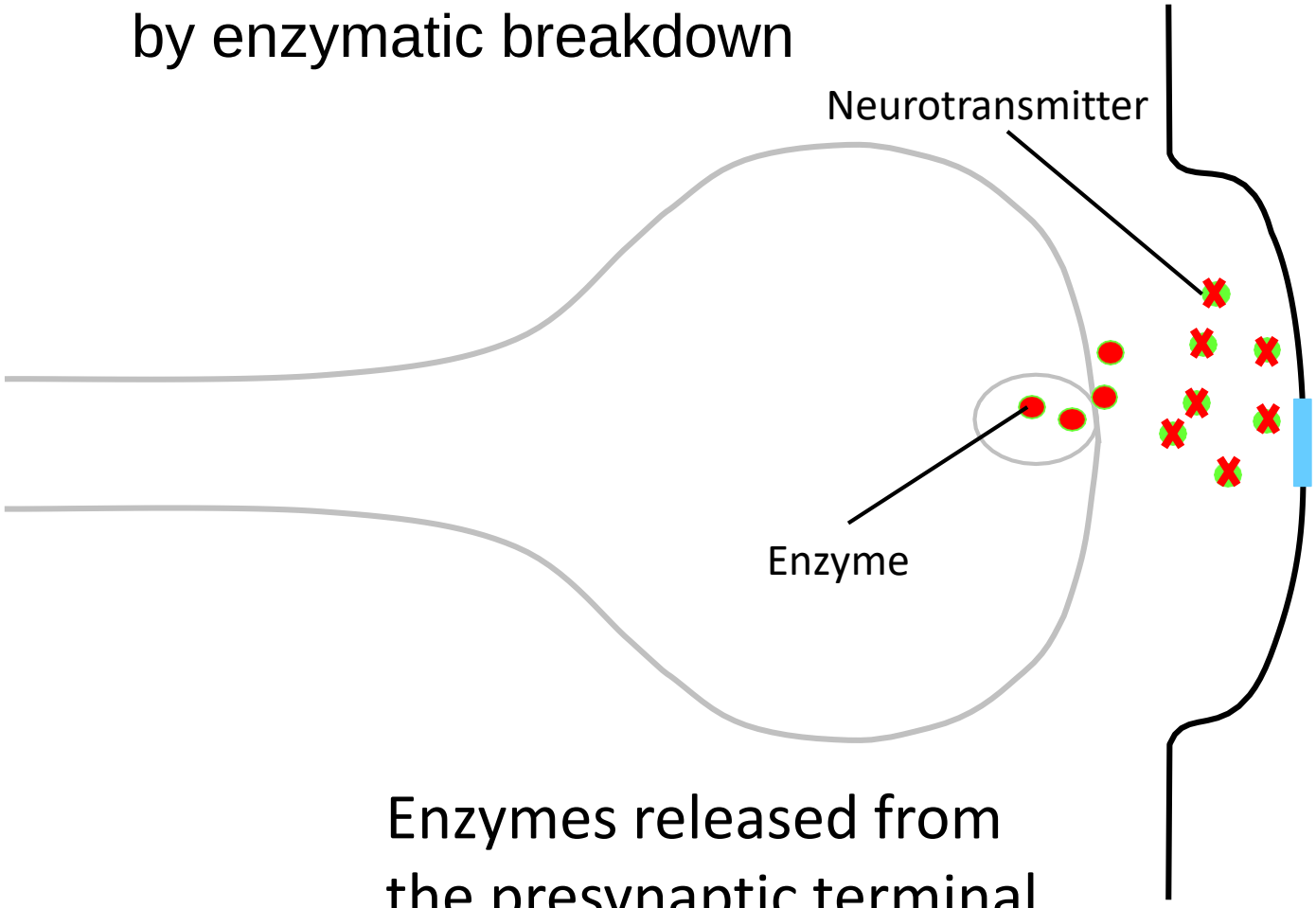


Neurotransmitter is repackaged into vesicles for reused

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Signal termination

Rapid termination of signal
by enzymatic breakdown



Enzymes released from
the presynaptic terminal

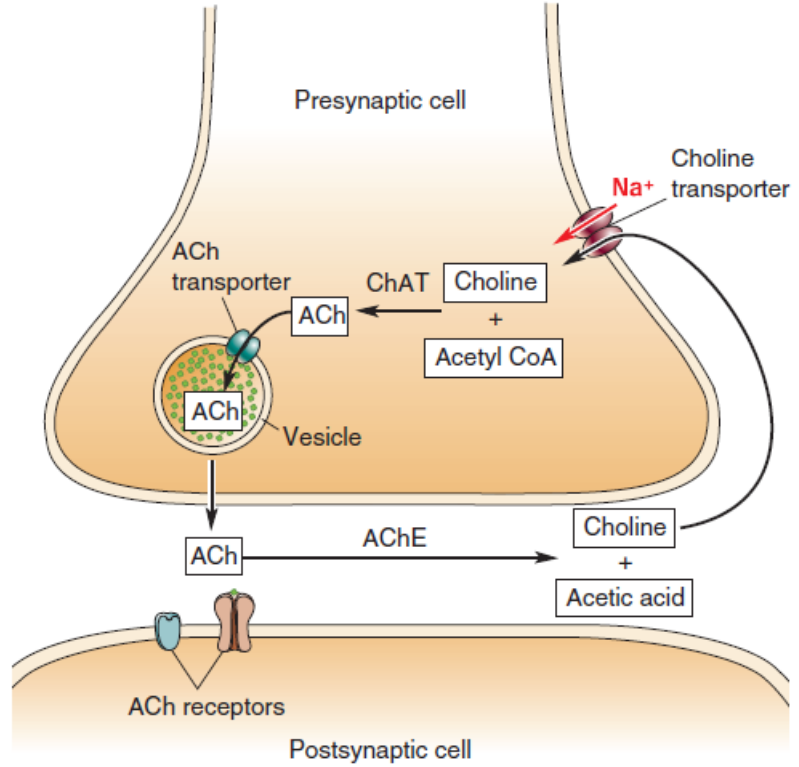


FIGURE 6.10
The life cycle of ACh.

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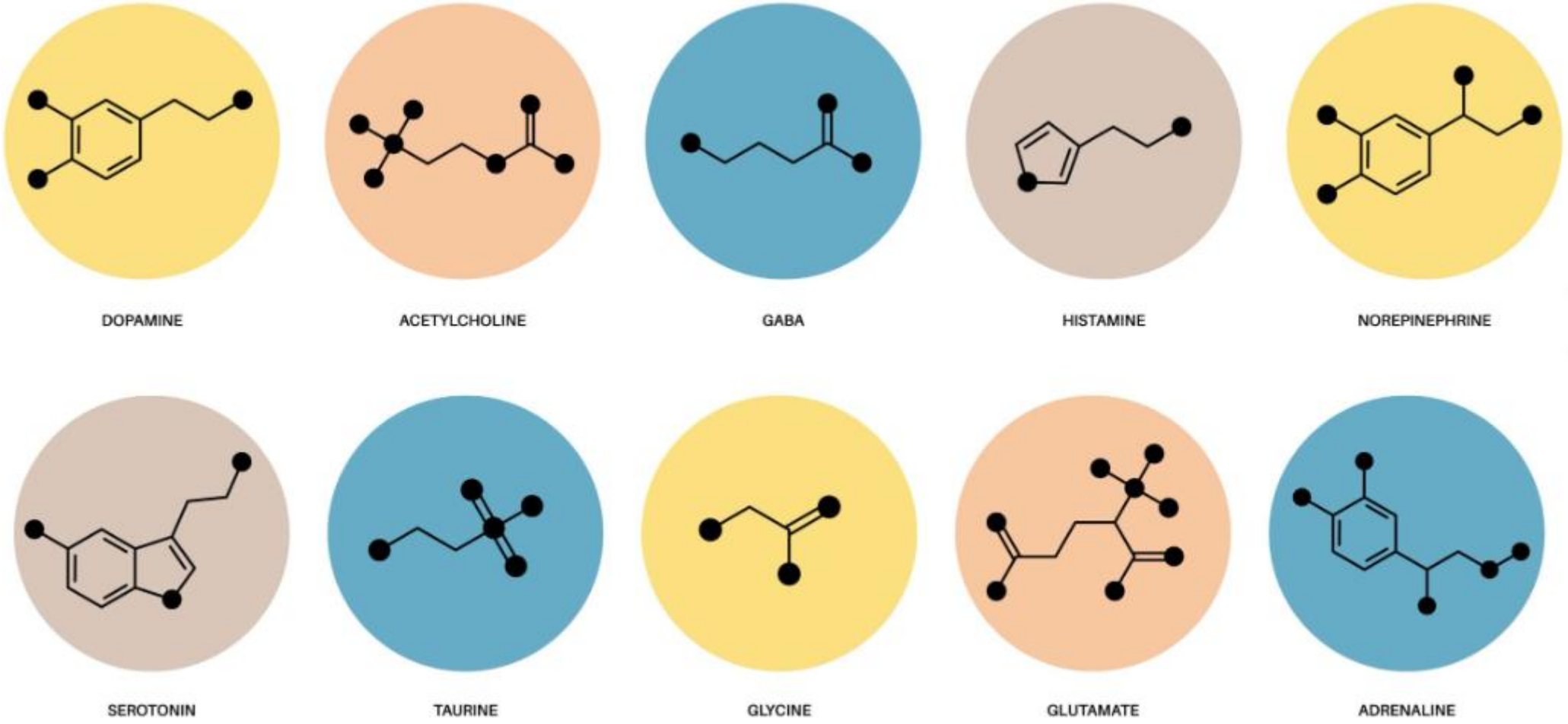
Neurotransmission summary

Neurotransmission is a **stepwise process**:

- **Action potential** arrives at the **axon** triggers the release of **neurotransmitters** into the **synaptic cleft**
- **Neurotransmitters** bind to the receptors on the **postsynaptic** neuron triggering a response
- **Termination** of the signal requires the removal of the neurotransmitter via:
  - **Reuptake pumps** - Glutamate transporters for glutamate
  - **Enzymatic degradation** - Acetylcholinesterase that breaks down acetylcholine

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Neurotransmitters - *chemical messengers at the synapse*



The major neurotransmitters

Amino acids: - Glutamate (major excitatory)
- Gamma (γ)-aminobutyric acid (GABA); (major inhibitory)
- Glycine (inhibitory)

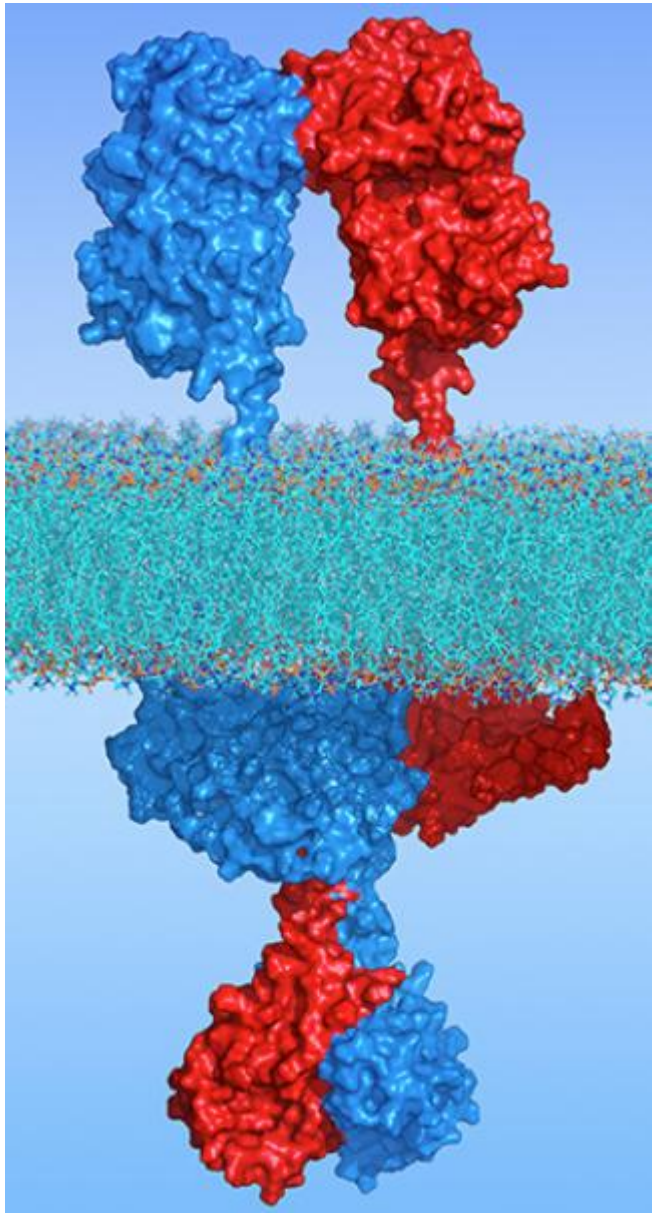
Monoamines: Noradrenaline, Dopamine, Serotonin

Further details in Synaptic Signalling Lecture

Acetylcholine (ACh): Major player, both excitatory and inhibitory

Neuroactive peptides:
>100 different kinds
e.g. Opioid peptides (endorphin, enkephalin)
Tachykinins (Substance P, neurokinin A)

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news.weill.cornell.edu

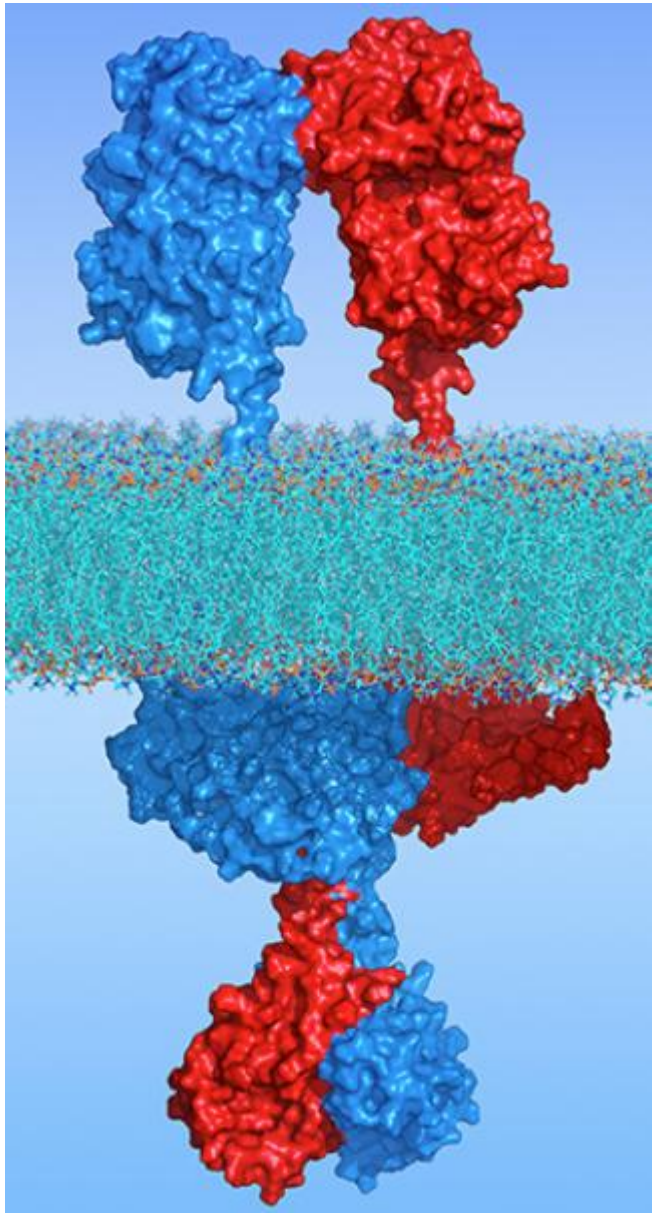
Receptors

Recognition site for a neurotransmitter

- initiates the intracellular signal

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Receptors



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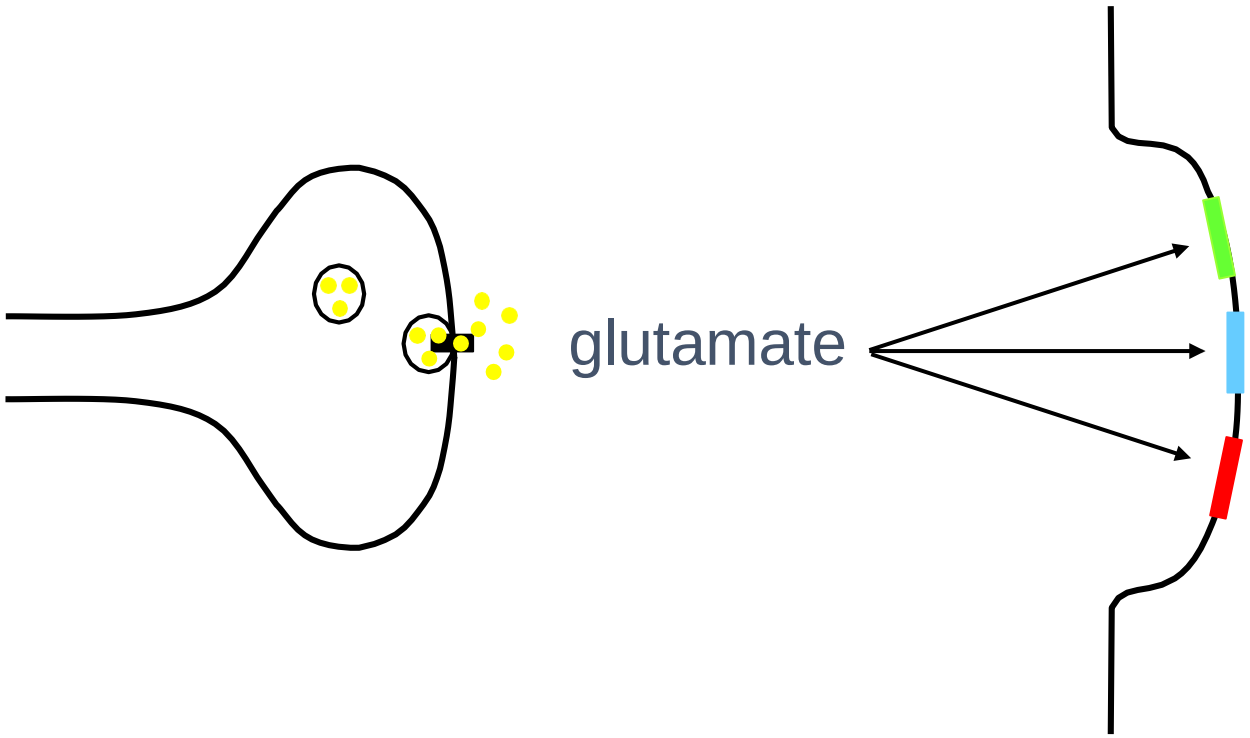
- *Membrane spanning protein molecules*
- Transmitter binding causes structural change ⇒ signal
- Receptors are specific for a neurotransmitter
- One neurotransmitter - several receptor subtypes

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Receptor nomenclature

Pre-synaptic

Post synaptic

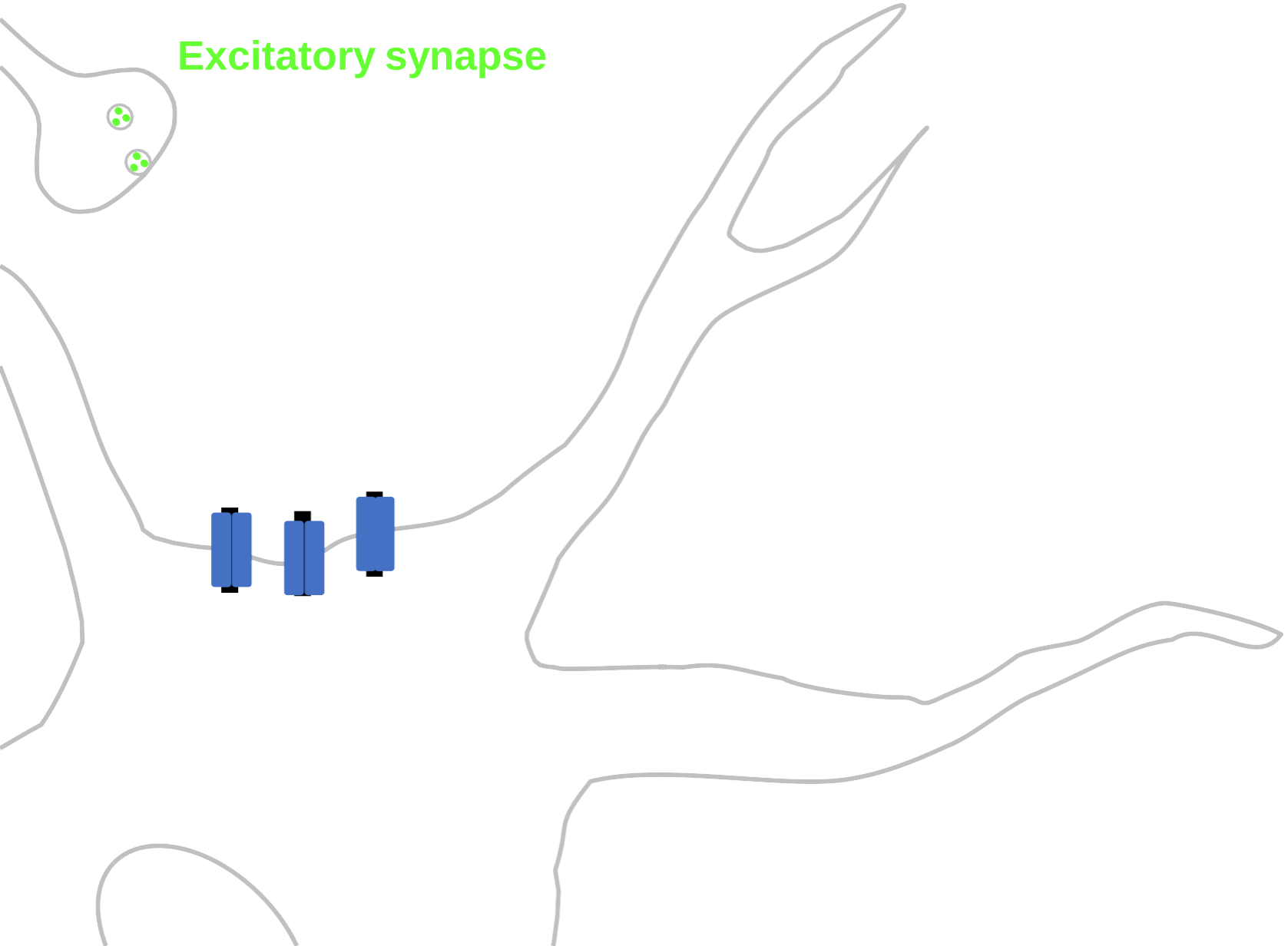


multiple receptor sub-types

Two major **superfamily** based on how they work **ionotropic receptors** and **metabotropic receptors**

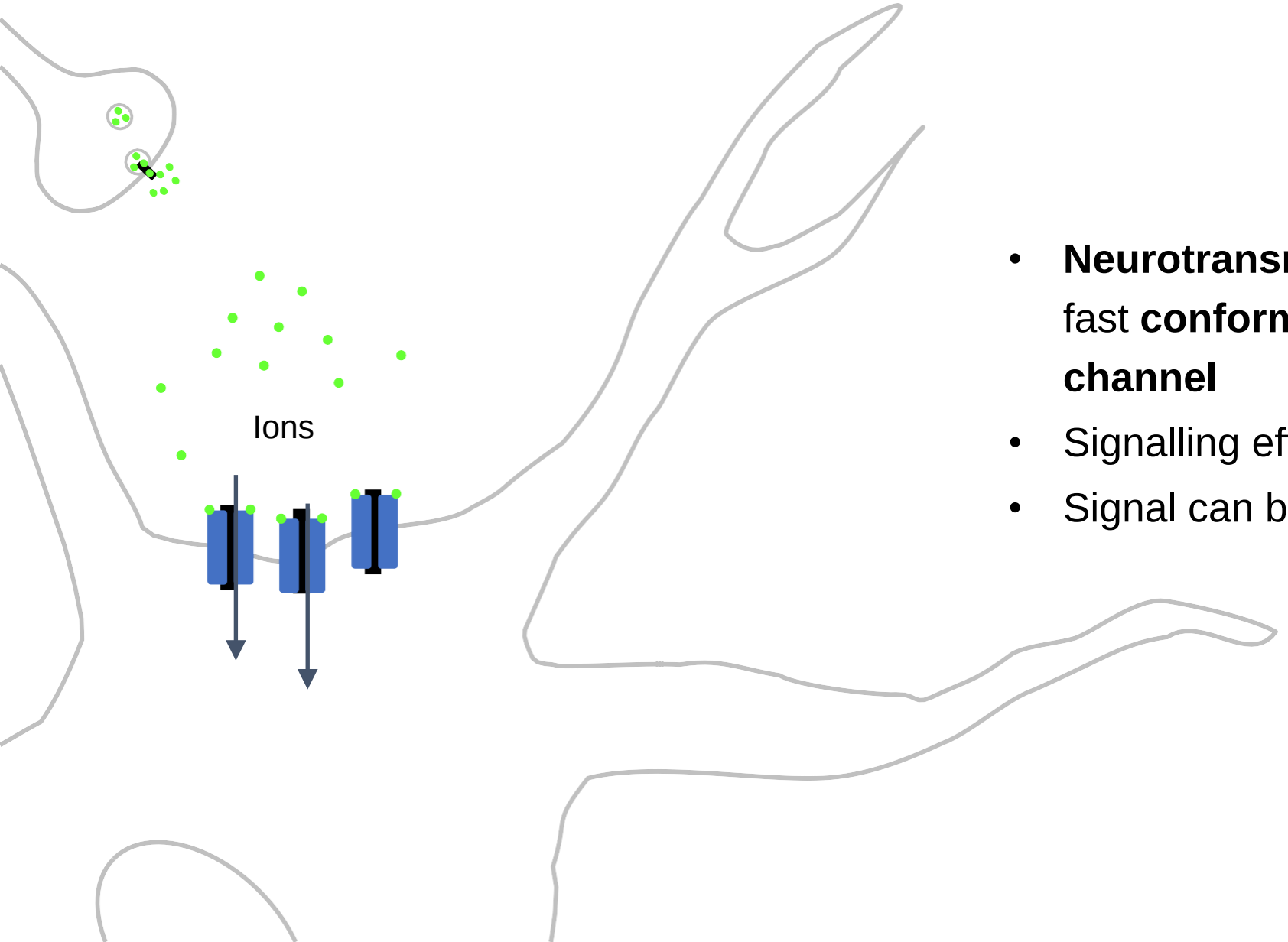
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Ionotropic receptor



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Signalling by ionotropic receptor

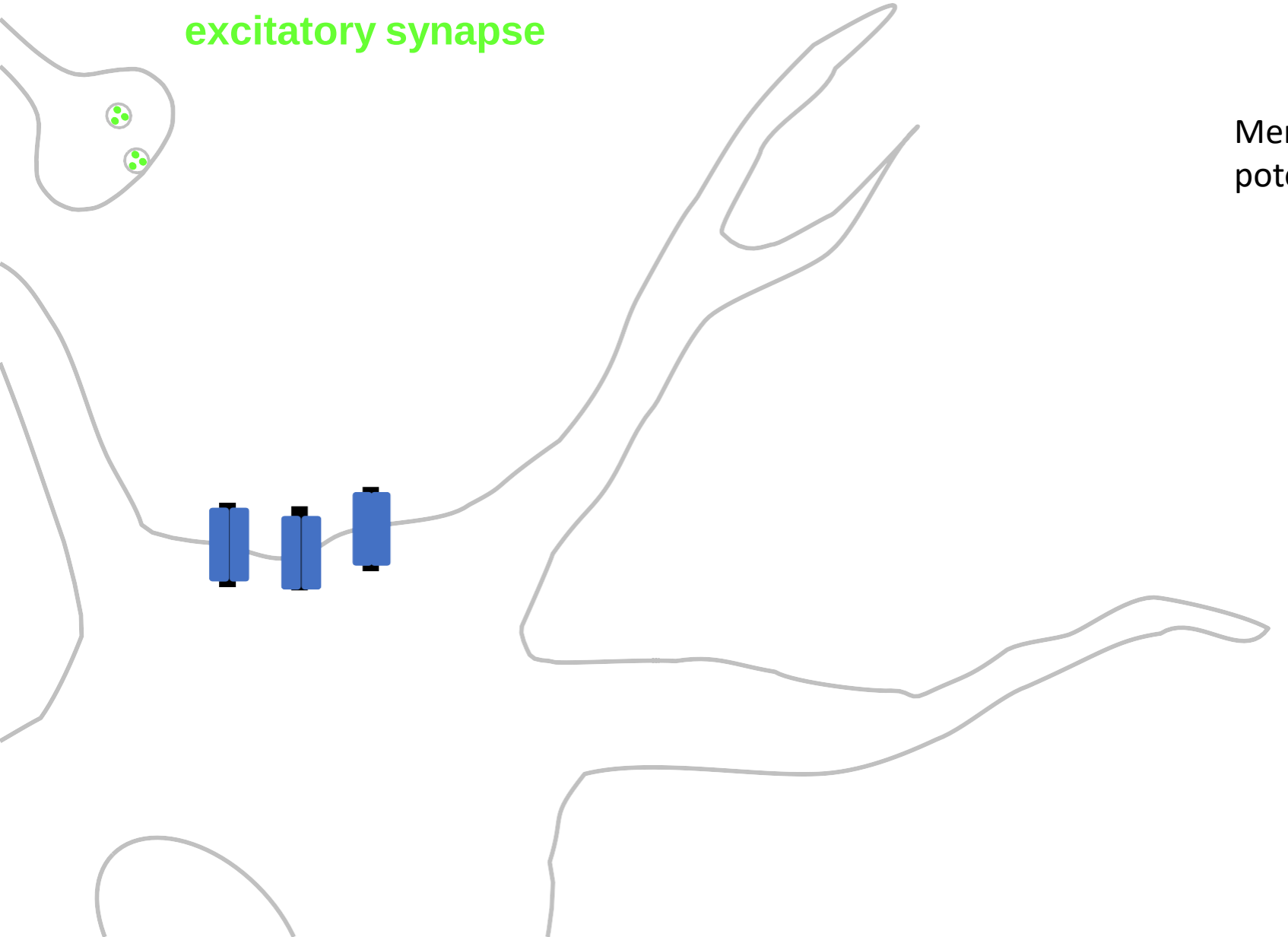


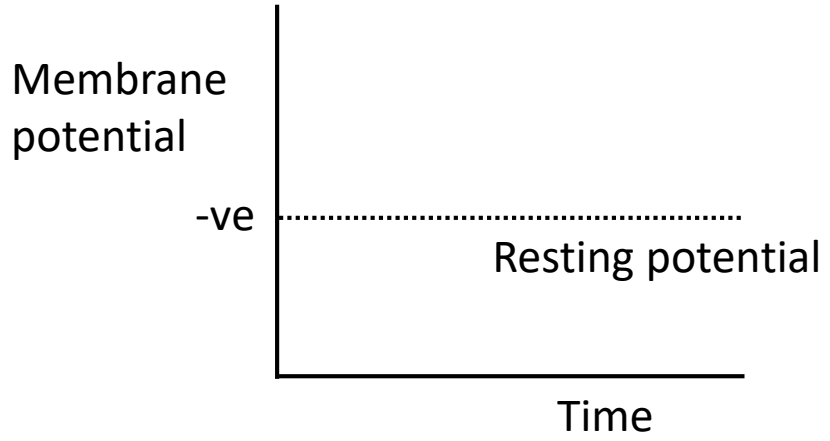
- **Neurotransmitter** binds to receptor inducing a fast **conformational change** that opens an **ion channel**
- Signalling effect is fast
- Signal can be **excitatory** or **inhibitory**

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Activation via ionotropic receptor

excitatory synapse

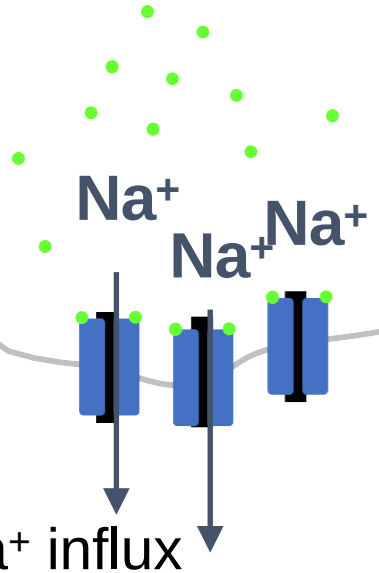




An **unstimulated** neuron is **negatively** charged

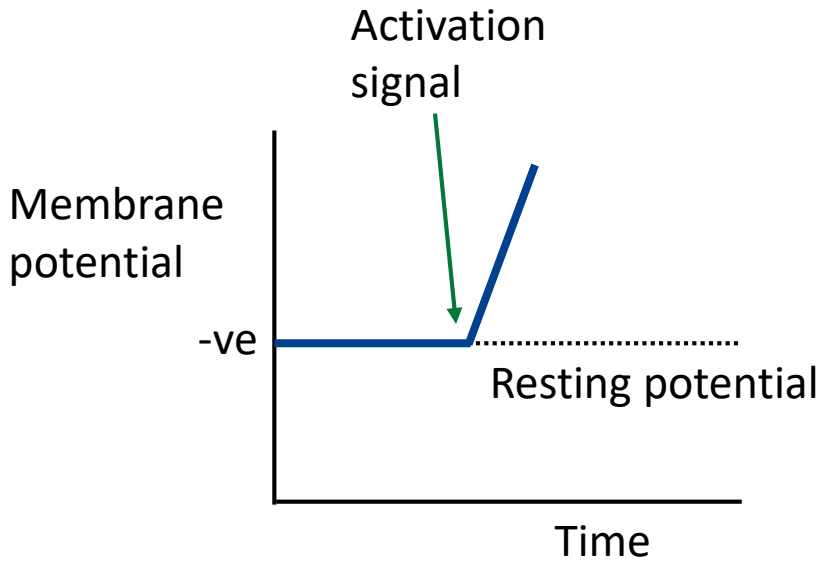
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Signalling by ionotropic receptor



Na^+ influx

⇒ **membrane depolarisation**



Excitatory ionotropic receptor

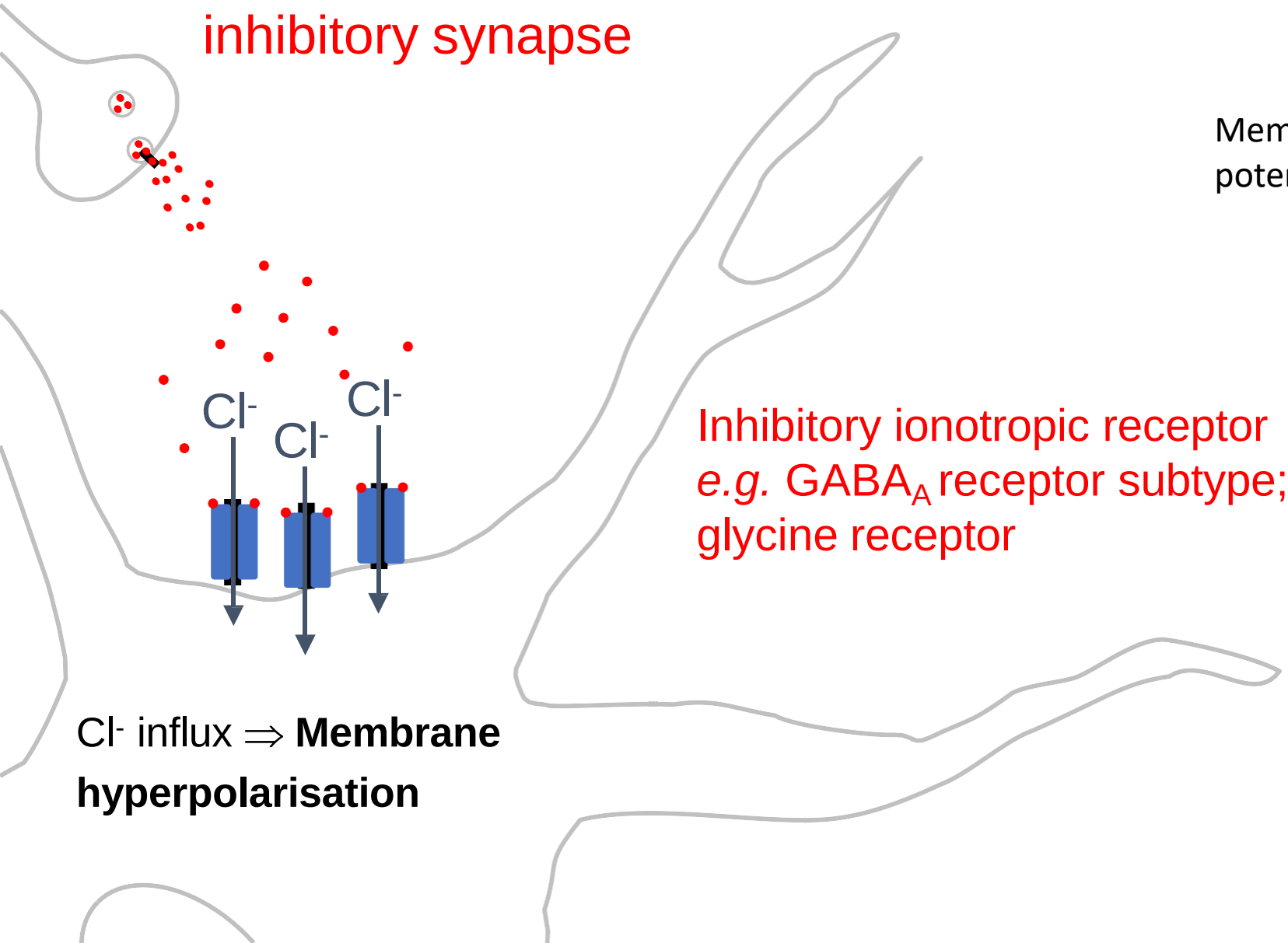
e.g. glutamate AMPA, NMDA, kainate receptor subtypes

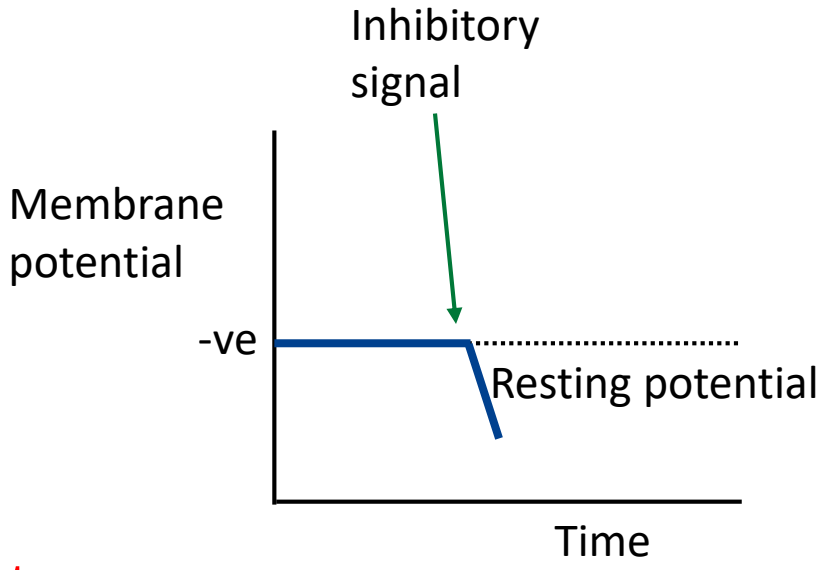
ACh nicotinic receptor subtype

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Signalling by ionotropic receptor

inhibitory synapse





Inhibitory ionotropic receptor
e.g. $GABA_A$ receptor subtype; glycine receptor

Cl^- influx ⇒ **Membrane hyperpolarisation**

When inhibited, Cl^- ions flow into the cell to cause membrane to hyperpolarise (become more –ve), making it more difficult to become activated

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Receptor signalling mechanisms

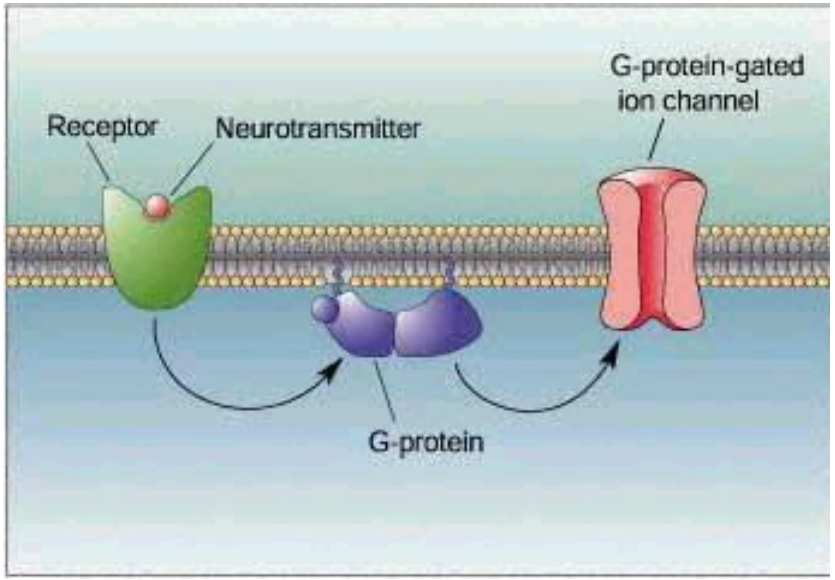
1. Ionotropic

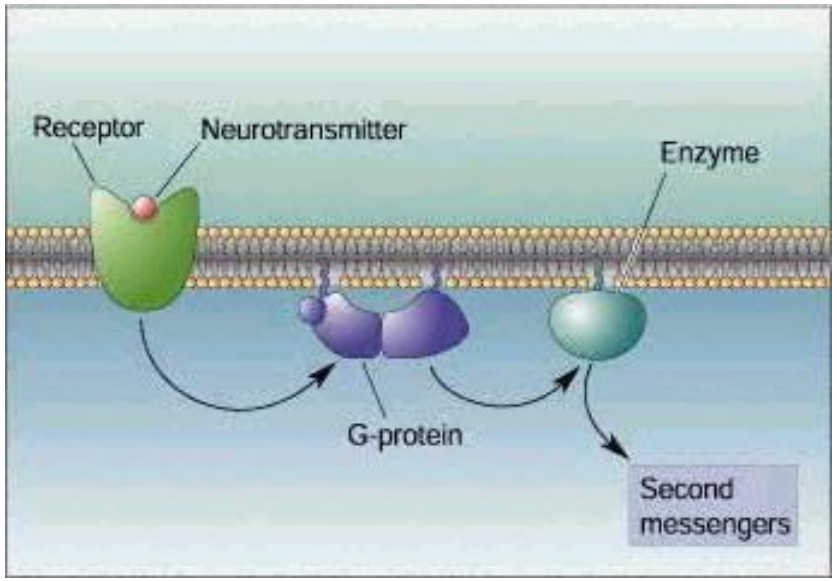
- Receptor operated/ligand gated channels

- Transmitter binding ⇒ Conformational change
  ⇒ Channel opening
  ⇒ Ion movement
  Fast transmission

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Metabotropic (G-protein coupled) receptor





Activated G-protein can:

- Open or close ion channels
- Stimulate or inhibit enzymes/

  secondary messenger systems

Further details in Concepts of Brain Function, Toxins and Medicines Lectures

e.g. Ach muscarinic; $GABA_B$; monoamine receptors (except $5\text{-}HT_3$)

*can have fast and slow transmission by same neurotransmitter

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'Neuroscience- Exploring the Brain' Bear, Connors, Paradiso

Receptor signalling mechanisms

2. Metabotropic

- G -protein coupled receptors

- Transmitter binding ⇒ Conformational change
  ⇒ Activates G-protein
  ⇒ Activates ‘effector systems’
  ⇒ Indirect effects on excitability

- ***Slower transmission***
- ***Longer lasting effects***

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Summary

- Overview of the nervous system
- How signals are transmitted between neuronal cells
  - Electrical synapse (Gap Junction)

    - Direct contact allows bi-directional communication
  - Chemical synapse

    - Action potential and Ca^{2+}-dependant release of neurotransmitter from pre-synaptic terminal $\Rightarrow$ receptor on post synaptic element

- Receptor nomenculature, signalling mechanisms and subtypes
  - Ionotropic receptor, direct opening of ion channel (fast)
  - Metabotropic receptor, triggers effector system thro' G-protein activation (slower)

- Ion fluxes control membrane polarisation (cell excitability)

  - Na^+ influx $\Rightarrow$ depolarisation

  - Cl^- influx $\Rightarrow$ hyperpolarisation

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Further Reading

- Bear, Connors, Paradiso *'Neuroscience- Exploring the Brain'*
- Berne & Levy *'Principles of Physiology'*
- *Both text books are available online at Birmingham Library*

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Foundations of Neuroscience (FoN)

Practice questions



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Question 1

Which of the following is the central nervous system (CNS)?

A. Brain and peripheral nerves

B. Brain and spinal cord

C. Spinal cord and sensory neurons

D. Cranial nerves and spinal nerves

E. Brain and autonomic nervous system

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Question 2

Which of the following correctly describes the sympathetic nervous system?

A. It controls voluntary skeletal muscle movement

B. It is primarily responsible for "rest and digest" functions

C. It is associated with "fight or flight" responses

D. It produces cerebrospinal fluid

E. It provides sensory input to the brain

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Question 3

Which glial cell is primarily responsible for insulation and myelination to improve signal transmission?

A. Astrocyte
B. Microglia
C. Ependymal cell
D. Oligodendrocyte
E. Neuron

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Question 4

What is the primary function of microglia?

A. Production of cerebrospinal fluid

B. Myelination of neurons

C. Metabolic support of neurons

D. Immune surveillance and phagocytosis

E. Generation of action potentials

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Question 5

Which neuronal structure is primarily responsible for receiving input from other neurons?

A. Axon terminal

B. Axon hillock

C. Dendrite

D. Synaptic cleft

E. Myelin sheath

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Question 6

What is the primary role of the axon hillock?

A. Generation of the action potential

B. Reception of neurotransmitters

C. Production of cerebrospinal fluid

D. Release of neurotransmitters

E. Degradation of neurotransmitters

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Question 7

Which type of neuron is described as having one axon and many dendrites and is the most common neuronal type in humans?

A. Unipolar
B. Bipolar
C. Pseudo-unipolar
D. Multipolar
E. Monopolar

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Question 8

Which statement best describes an electrical synapse?

A. Neurotransmitters diffuse across a synaptic cleft

B. It is usually unidirectional

C. It involves direct ion flow through gap junctions

D. It is slower than chemical transmission

E. It depends on neurotransmitter reuptake

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Question 9 (SAQ)

Compare the functions of neurons and glial cells in the brain.

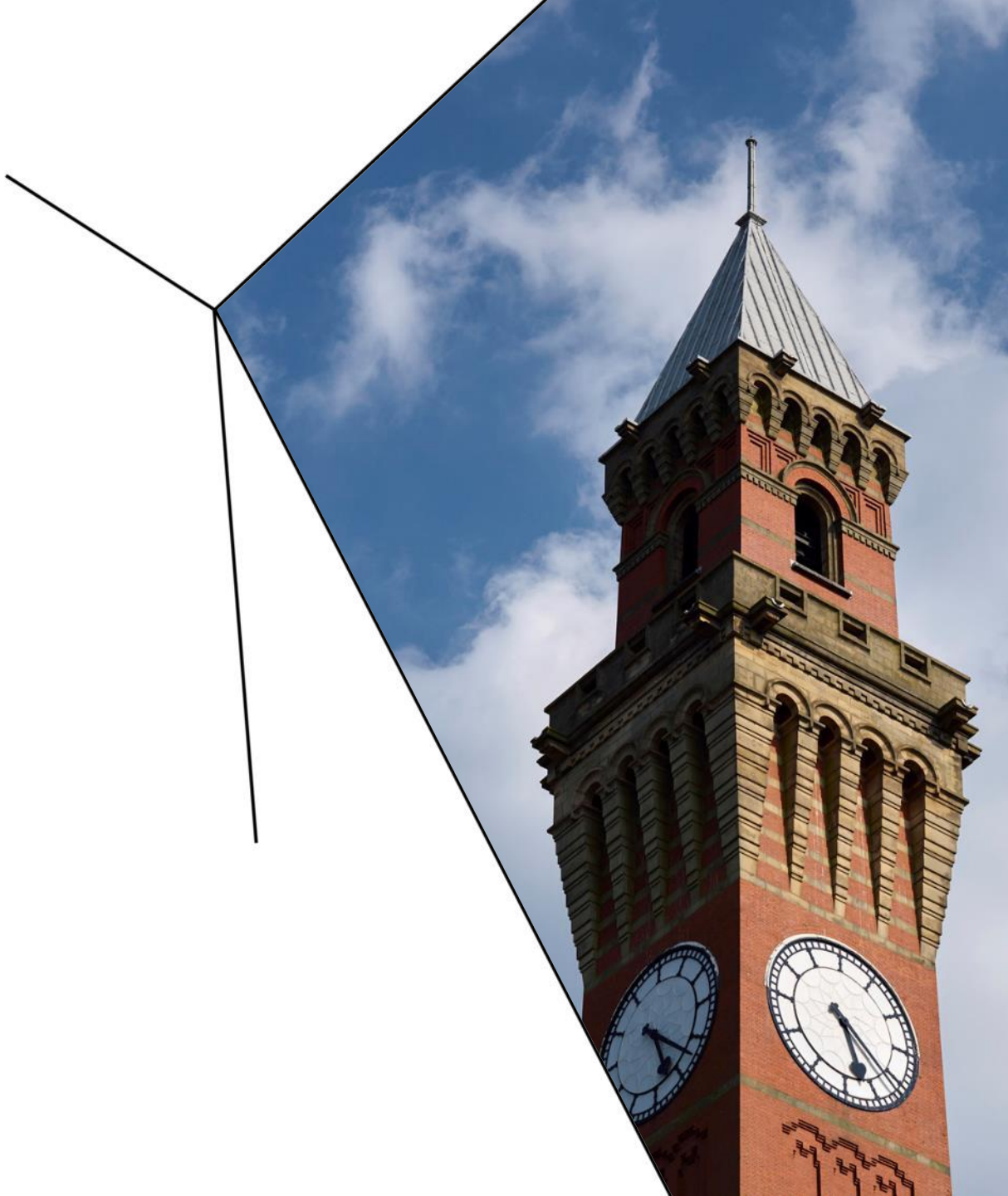
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Question 10 (SAQ)

Describe the specialised functions of the major regions of a neuron involved in communication.

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Questions



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